

EEE 402 (January 2026)
Artificial Intelligence and Machine Learning Laboratory

Final Project Report

Section: G2 Group: 13

Telling Lung Adenocarcinoma from Squamous Cell
Carcinoma on CT Using Hospital-Robust Radiomics

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1 Abstract

Adenocarcinoma (ADC) and squamous cell carcinoma (SCC) are the two major types of non-small cell lung cancer and are treated differently. The type is confirmed today by biopsy, which is invasive, but almost every patient already has a chest CT scan. In this project we developed a fully automatic CT pipeline predicting ADC vs SCC, and we designed the evaluation of it such that the result holds at hospitals the model has never seen.

We first demonstrated that pooling CT data from different hospitals results in a shortcut. Hospitals differ in scanner style and ADC / SCC mixture, and a predictor with access to the hospital name only obtains a pooled ROC-AUC of 0.71 without looking at any image. To mitigate this effect we used a site-balanced AUC, ComBat harmonisation fitted within each training fold and leave-one-hospital-out testing. The expert box or TotalSegmentator finds the tumour and MedSAM2 segments it. Three peritumoral shells (0-4, 4-8 and 8-12 mm) are used to extract texture features and combined with tumour position, size, age and sex. Each of the two feature blocks is modelled by an ensemble of logistic regression, SVM, LightGBM and ridge regression and the block scores are fused.

The model achieved a site-balanced AUC of 0.722 (95% CI 0.687-0.762) on 1041 patients from four public cohorts (697 ADC, 344 SCC) in repeated nested cross-validation and an average of 0.714 for an unseen hospital. In two external tests (locked and pre-registered) it scored 0.686 on Lung3 (79 patients) and 0.585 on TCGA (84 patients). Thin-slice CT performed best for external results and 5 mm CT was the main failure mode. For this data size, larger deep models (attention-MIL, CNN features, CT foundation models and adversarial fine-tuning) did not outperform the simpler harmonised radiomics model.

Keywords: lung cancer, histologic subtype, radiomics, ComBat harmonisation, multi-centre validation, site-balanced AUC, MedSAM2.

2 Introduction

Lung cancer is the most common cause of cancer mortality worldwide [1]. About 85% of the cases are non-small cell lung cancer (NSCLC) and its two most common histologic types are adenocarcinoma (ADC) and squamous cell carcinoma (SCC) [2]. It is important to know the type as this will affect treatment. Some chemotherapy drugs are for one type and not the other and most targeted treatments (for example for EGFR and ALK changes) are relevant for ADC mainly. Currently biopsy is the only way to confirm the type: a needle through the chest wall, a bronchoscope or surgery. Biopsy is invasive, carries risks such as collapsed lung and bleeding, and sometimes the sample is too small to make a decision.

Almost all patients with lung cancer undergo a chest CT scan before any treatment. The CT shows different tendencies of ADC and SCC. ADC is more often peripheral, and may have fuzzy or ground-glass edges. SCC is more often central, solid and larger, and is more common in men and heavy smokers. These are tendencies, not rules, and so these two types often look alike. Radiomics attempts to capture such differences by calculating many quantitative features (intensity, texture and shape) from the tumour region and feeding them into a machine-learning model [3, 4]. So our project question was: can a computer tell ADC from SCC if we only give it the CT scan, and does the answer still hold up at hospitals not used for training?

The project was focused on the second half of this question. Public CT collections are from different hospitals, each with their own scanners and their own mix of patients. One of the hospitals in our data has 75% SCC, while the other three are 73%-81% ADC. A model trained on the pooled data can learn to recognise the hospital from scanner texture and then guess the majority class of that hospital. This is a well-known kind of shortcut learning [5, 6]. It scores well on pooled data but fails at a new hospital. Much of this work was spent measuring this shortcut and building an evaluation that it cannot fool.

Objectives

The project's objectives were:

1. to develop a fully automated CT pipeline (tumour detection, outlining, feature extraction and classification) for ADC vs SCC;
2. to measure the hospital shortcut and to use a metric and validation design that are not affected by it;
3. to reduce scanner differences between hospitals by harmonisation, fitted on training data only;
4. to compare simple handcrafted models with deep learning counterparts under the same honest protocol;
5. to test the frozen model on two external datasets which were locked until the end.

Why this is a Complex Engineering Problem

Table 1 shows the mapping of the project to the attributes of a complex engineering problem. The main difficulty was that there was no obvious solution: a higher score on pooled data can be a worse model, so success itself had to be defined carefully.

Attribute	How it appears in this project
P1: Depth of knowledge	Needs CT physics (Hounsfield units, slice thickness, reconstruction kernels), image processing (resampling, segmentation), statistics (empirical Bayes, bootstrap) and machine learning.
P2: Conflicting requirements	Higher pooled accuracy conflicts with fairness across hospitals; removing scanner effects can also remove real disease signal; model capacity conflicts with small data.
P3: Depth of analysis	No closed-form answer. Many design choices (regions, features, harmonisation, models, fusion) had to be tested in nested cross-validation without leaking test information.
P4: Familiarity of issues	Hidden confounding between hospital and label is not visible in standard metrics and is rarely checked in the literature.
P5: Extent of applicable codes	Radiomics standards (IBSI), DICOM, data licences (CC BY, CC BY-NC) and good practice for medical AI evaluation (pre-registration, locked test sets).
P6: Stakeholders	Patients, radiologists, pathologists, oncologists and hospitals with different scanners.
P7: Interdependence	Mask quality affects features, features affect harmonisation, harmonisation affects model selection; a change in one stage changes all later stages.

Table 1: Mapping of the project to the attributes of a complex engineering problem

Possible alternative solutions

There are several other routes that could meet the same clinical need. Table 2 compares them to the one we picked.

Alternative	Advantage	Disadvantage
Biopsy only (current practice)	Gold standard; also gives tissue for gene tests	Invasive, risky, sometimes not diagnostic, slow
Visual reading by a radiologist	No new tools needed	Subjective; ADC and SCC often look alike
End-to-end deep CNN on CT	Learns features automatically	Needs thousands of patients; tends to learn scanner style (shown in Section 5.3)
Single-hospital radiomics	Simple, high internal scores	Rarely transfers to other hospitals
PET/CT based model	Uses metabolic information	PET is not available for every patient
Multi-hospital harmonised radiomics (chosen)	Works with routine CT, small models, honest multi-centre evaluation	Moderate accuracy; sensitive to thick CT slices

Table 2: Other solutions considered

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of the project was set after the progress presentation (Week 10). In scope:

- binary classification of ADC vs SCC from a pre-treatment chest CT scan along with the patient's age and sex;
- public, de-identified CT collections from The Cancer Imaging Archive (TCIA) [7] and the NCI Imaging Data Commons (IDC) [8];
- fully automatic tumour localisation and outlining so that the same method works in a new hospital;
- evaluation within the development hospitals (cross-validation), on unseen development hospitals (leave-one-hospital-out) and on two locked external datasets.

Not included: other NSCLC types (large cell, not otherwise specified), PET images, prediction of gene mutation, prediction of survival, any clinical deployment. The model is a research decision support tool, not a substitute for biopsy.

3.1.2 Literature Review

Radiomics for histologic subtype

Zhu *et al.* [9] selected five radiomic features using LASSO from 129 patients in a single hospital and reported an AUC of 0.91 but without external testing. Yang and colleagues [10] evaluated 645 patients from three centres. AUC was about 0.78 when centres were mixed randomly, but dropped to 0.54-0.64 when training on one centre and testing on another. Pasini *et al.* [11] used LUNG1 and NSCLC-Radiogenomics (2 of our cohorts) and found the accuracy to drop from 0.77 to 0.59 after ComBat harmonisation, indicating that some of the pooled performance was due to dataset differences. Song *et al.* [12] collected 868 patients from eight TCIA databases and reported AUCs of 0.82 internally and 0.82 and 0.80 on TCGA and Lung3, selecting the best of 130 radiomics models, and using expert tumour masks.

Deep learning

Chaunzwa *et al.* [13] trained a CNN on 311 patients from a single hospital (internal AUC 0.71) and achieved 0.60 on an external Lung3 subset. Chen *et al.* [14] used a multi-task CNN on 402 TCIA patients (internal 0.84, external 0.73). General-purpose image features from foundation models pre-trained on large CT collections, such as FMCIB [15] and CT-FM [16], can be reused without training a network from scratch.

Peritumoral and local information

Tang *et al.* [17] showed that features from the region around the tumour add information to the features from inside it. Vuong *et al.* [18] used local radiomic activation maps to see where the signal lies inside and around the tumour. Our first idea for the project was to use weakly supervised attention-based multiple instance learning (MIL) [19, 20], which learns from many local patches when only a patient-level label is available.

Multi-centre effects

ComBat [21] was originally developed to remove batch effects from gene expression data, and was adapted to radiomics [22]. Shortcut learning of hospital identity has been documented in chest radiographs [5], and in deep networks generally [6]. The IBSI [23] standardises definitions of radiomic features, but standard definitions do not eliminate scanner differences.

Gap

Most published ADC/SCC models report high scores from a single hospital or from random splits of pooled data. Very few measure how much of the score is explained by the hospital itself. Very few test a model that was frozen before opening the external labels. Our project fills this gap.

3.1.3 Formulation of Problem

We have N patients from H hospitals. Patient i has a CT volume I_i , a hospital label $h_i \in \{1, \dots, H\}$ and a histology label $y_i \in \{0,1\}$. $y_i = 1$ means ADC (the positive class), and $y_i = 0$ means SCC. We want a function f that takes a CT scan and outputs a real-valued score $s_i = f(I_i)$, such that $\hat{y}_i = \mathbb{1}[s_i > 0]$. We quantify ranking quality with the ROC-AUC, the probability that a randomly chosen ADC patient has a higher score than a randomly chosen SCC patient:

$$\text{AUC} = \frac{1}{n_1 n_0} \sum_{i: y_i=1} \sum_{j: y_j=0} [\mathbb{1}(s_i > s_j) + \frac{1}{2} \mathbb{1}(s_i = s_j)] \quad (1)$$

where n_1 and n_0 are the number of ADC and SCC patients, respectively. The real target is not the AUC on pooled data, but the AUC at a hospital h^* that was not used for training:

$$f^* = \underset{f}{\operatorname{argmax}} \mathbb{E}_{h^* \notin \mathcal{H}_{\text{train}}} [\text{AUC}_{h^*}(f)] \quad (2)$$

From this goal we derived the following design requirements:

- R1. The model selection metric should assign 0.5 to any predictor that knows only the hospital.
- R2. All fitted steps (imputation, harmonisation, scaling, feature selection, hyperparameters) should be fitted only on the training patients.
- R3. Tumour masks must be generated automatically and in the same way in each hospital, so that the mask source cannot become a new hospital fingerprint.
- R4. We need to process a new hospital without labels.
- R5. The model should be small enough to train on a laptop GPU (4 GB) and to avoid overfitting on about 1000 patients.
- R6. External tests have to be pre-registered, frozen with a hash and scored once.

3.1.4 Analysis

Figure 1 shows the ADC/SCC mix for each cohort. LUNG1 is 25% ADC, versus 79%, 81% and 73% ADC for Radiogenomics, Lung-PET-CT-Dx and NLST respectively. Consider a predictor that ignores the image entirely and only outputs the ADC rate of the patient's hospital:

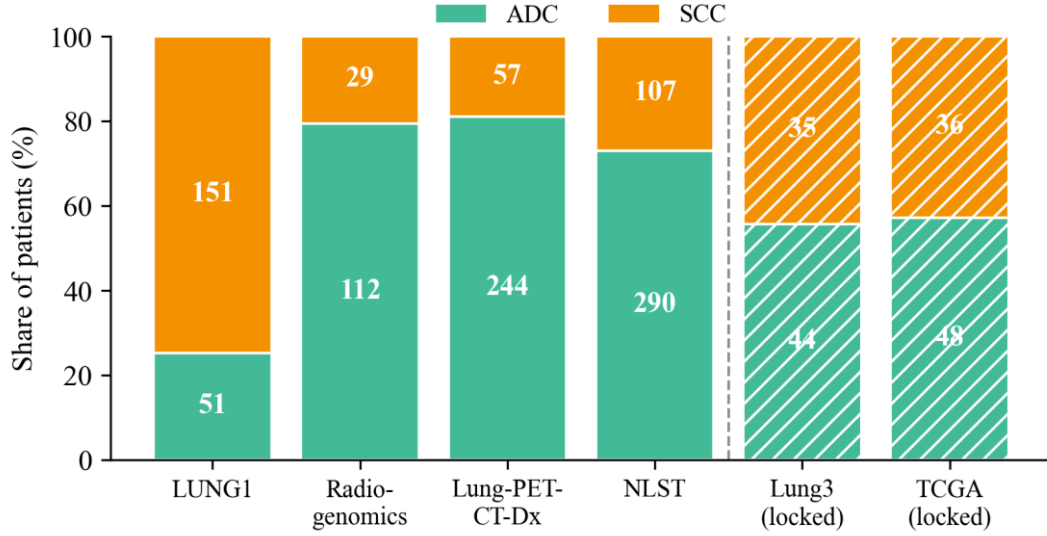


Figure 1: Class mix of the four development cohorts and the two locked external test sets. Numbers in the bars are numbers of patients. LUNG1 is flipped relative to the other hospitals.

$$s_i = \pi_{h_i}, \pi_h = \frac{n_{h,1}}{n_{h,1} + n_{h,0}} \quad (3)$$

All ADC patients from a high ADC hospital are ranked above all SCC patients from LUNG1. This predictor has a pooled AUC of 0.710 on our 1041 patients (Figure 2). The score of a naive radiomics model is 0.727, which is only slightly better than knowing the hospital. A pooled AUC of about 0.7 therefore tells us very little about whether a model has learned anything about the tumour.

To satisfy requirement R1 we weight each patient with the inverse size of its (hospital, class) group:

$$w_i = \frac{1}{H \cdot n_{h_i, y_i}} \quad (4)$$

and define site-balanced AUC as the weighted version of Eq. (1):

$$\text{AUC}_{\text{SB}} = \frac{\sum_{i: y_i=1} \sum_{j: y_j=0} w_i w_j [\mathbb{1}(s_i > s_j) + \frac{1}{2} \mathbb{1}(s_i = s_j)]}{\sum_{i: y_i=1} w_i \sum_{j: y_j=0} w_j} \quad (5)$$

The groups $2H$ (hospital, class) have the same total weight $1/H$ after weighting. For the hospital-only predictor of Eq. (3) pairs from the same hospital are ties, contributing $1/2$. For two different hospitals h and h' , (ADC from h , SCC from h') and (ADC from h' , SCC from h) together make 1 exactly, whichever of π_h and $\pi_{h'}$ is bigger. Therefore

$$\text{AUC}_{\text{SB}}(\pi) = \frac{1}{H^2} \left[\sum_{h=1}^H \frac{1}{2} + \sum_{h < h'} 1 \right] = \frac{1}{H^2} \left[\frac{H}{2} + \frac{H(H-1)}{2} \right] = \frac{1}{2} \quad (6)$$

so the shortcut is worth exactly nothing according to this metric, as desired. On the real data the hospital-only predictor scores 0.497 and the naive radiomics model 0.642 (Figure 2). Training uses the same weights w_i as sample weights, so the classifiers are not rewarded for predicting the hospital either. The second part of the analysis is on scanner style: a logistic regression trained to recognise the hospital from texture features achieved an AUC of 0.81-0.91 (Section 5.3.3). This motivated harmonisation (Section 3.2.4).

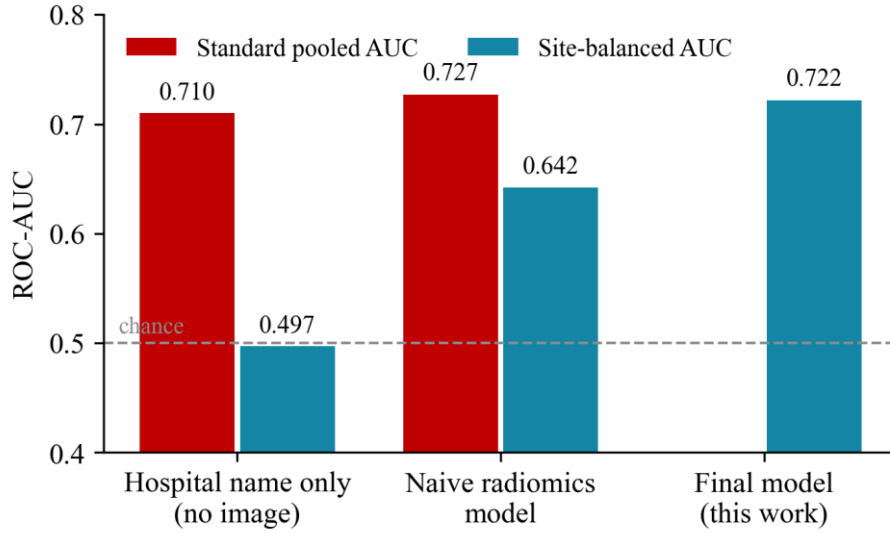


Figure 2: The hospital shortcut. A predictor with knowledge of only the hospital has a pooled AUC of 0.710 but is exactly at chance under the site-balanced AUC. The final model achieves 0.722 under the fair metric.

3.2 Design Method (PO(a))

This section presents the mathematics for each stage of the pipeline. The overall flow is shown later in Figure 4.

3.2.1 Image Preprocessing

Each DICOM series is stacked into a single 3D volume and converted to NIFTI. Because scans have pixel sizes of 0.6-1 mm and slice gaps of 0.6-10 mm, all volumes are resampled to an isotropic grid of $2 \times 2 \times 2$ mm. The CT is linearly interpolated: the value at a new voxel location x is a weighted average of the 8 neighbouring original voxels,

$$I'(x) = \sum_{k=1}^8 w_k(x) I(x_k), \sum_{k=1}^8 w_k(x) = 1 \quad (7)$$

where the weights are products of the fractional distances along the axes. Masks are resampled to nearest neighbour to keep them binary. The intensities are clipped to the range between air and soft tissue. Gray levels are grouped into bins of width $\Delta = 20$ HU before texture is computed:

$$\tilde{I}(x) = \min(\max(I'(x), -1024), 200), b(x) = \lfloor \frac{\tilde{I}(x) + 1024}{\Delta} \rfloor + 1 \quad (8)$$

Clipping removes the effect of bone and metal and binning makes texture less sensitive on small noise. We chose 2 mm as a compromise: many scans are 3-5 mm slices and a finer grid would only invent detail.

3.2.2 Tumour Segmentation and Peritumoral Shells

The tumour location is an expert box for development cohorts (LUNG1 and Radiogenomics: bounding box of the expert outline; Lung-PET-CT-Dx and NLST: expert boxes). The box for external data is taken from the TotalSegmentator lung-nodule model [24], using the largest detected nodule. The box is given as a prompt to MedSAM2 [25], which is a medical version of the Segment Anything 2 model. MedSAM2 treats axial slices as frames in a video. It draws the outline of the tumour on the key slice and propagates the outline up and down. Segmentation quality against the expert outline B is evaluated using the Dice score,

$$\text{DSC}(A, B) = \frac{2|A \cap B|}{|A| + |B|} \quad (9)$$

which is 1 for identical masks and 0 for no overlap. From the tumour mask G we calculate the Euclidean distance of each voxel to the tumour and build three shells around it with radii $(r_0, r_1, r_2, r_3) = (0, 4, 8, 12)$ mm:

$$d(x) = \min_{g \in G} \|x - g\|_2, S_k = \{x: r_{k-1} < d(x) \leq r_k\} \quad (10)$$

The shells demarcate the tumour margin and its surroundings where ADC and SCC are expected to differ [17].

3.2.3 Radiomic and Position Features

Features are extracted from the original image with PyRadiomics [4], according to the definitions of the IBSI [23]. For each region, 93 texture features were extracted from six families: first-order statistics, GLCM, GLRLM, GLSZM, GLDM and NGTDM. The idea is shown in three examples. If $p(i)$ is the fraction of voxels in bin i , $p(i, j)$ the normalised grey-level co-occurrence matrix and $P(i, j)$ the number of runs of grey level i with length j then

$$H = - \sum_{i=1}^{N_g} p(i) \log_2(p(i) + \epsilon), \text{Contrast} = \sum_{i=1}^{N_g} \sum_{j=1}^{N_g} (i - j)^2 p(i, j) \quad (11)$$

$$\text{SRE} = \frac{\sum_i \sum_j P(i, j) / j^2}{\sum_i \sum_j P(i, j)} \quad (12)$$

Entropy H is a measure of randomness of intensity. GLCM contrast is a measure of local intensity jumps while short run emphasis (SRE) is large for fine textures. The final model has two blocks of features:

- Shells block (279 features): 93 texture features in each of the three shells.
- Position + clinical block (13 features): left or right lung; relative lateral, front-back and up-down position of the tumour centre; normalised distance from the midline; depth of the centre (in mm and relative); fraction of the tumour surface in contact with the lung boundary and with the medial (mediastinal) part of it; fraction of the tumour outside the lung; log tumour volume; age; sex.

Other blocks were tested as alternatives, including whole-tumour texture plus 14 shape features (GTV, 107 features), an inner 4 mm ring inside the tumour, FMCIB deep features [15] and CT-FM features [16].

3.2.4 Harmonisation with ComBat

ComBat [21] assumes that the hospital shifts and scales feature g of patient i in hospital h :

$$x_{ihg} = \alpha_g + \sigma_g \gamma_{hg} + \sigma_g \sqrt{\delta_{hg}} \varepsilon_{ihg}, \varepsilon_{ihg} \sim \mathcal{N}(0, 1) \quad (13)$$

where α_g is the overall mean, σ_g the pooled standard deviation, γ_{hg} the hospital location shift, and δ_{hg} the hospital scale factor. The raw shift and scale of each hospital after standardisation with pooled estimates are

$$z_{ihg} = \frac{x_{ihg} - \hat{\alpha}_g}{\hat{\sigma}_g}, \hat{\gamma}_{hg} = \frac{1}{n_h} \sum_{i \in h} z_{ihg}, \hat{\delta}_{hg} = \frac{1}{n_h} \sum_{i \in h} (z_{ihg} - \hat{\gamma}_{hg})^2 \quad (14)$$

Small hospitals give noisy estimates, so ComBat uses empirical Bayes. For all features of hospital h

it assumes $\gamma_{hg} \sim \mathcal{N}(\bar{\gamma}_h, \bar{\tau}_h^2)$ and $\delta_{hg} \sim \text{InvGamma}(\lambda_h, \theta_h)$, estimates these priors from the data and iterates the posterior means

$$\gamma_{hg}^* = \frac{n_h \bar{\tau}_h^2 \hat{\gamma}_{hg} + \delta_{hg}^* \bar{\gamma}_h}{n_h \bar{\tau}_h^2 + \delta_{hg}^*}, \delta_{hg}^* = \frac{\frac{1}{2} \sum_{i \in h} (z_{ihg} - \gamma_{hg}^*)^2 + \theta_h}{\frac{n_h}{2} + \lambda_h - 1} \quad (15)$$

until they converge. The harmonised feature is

$$x_{ihg}^* = \hat{\sigma}_g \frac{z_{ihg} - \gamma_{hg}^*}{\sqrt{\delta_{hg}^*}} + \hat{\alpha}_g \quad (16)$$

ComBat is fitted only on the training patients of each fold. A hospital that is not in the training data (an unseen or external hospital) gets its own $\hat{\gamma}$ and $\hat{\delta}$ from its unlabelled features with Eq. (14), so no test label is used. Figure 3 shows the effect on one real feature. Before ComBat the LUNG1 curve is clearly shifted; after ComBat the four hospitals overlap.

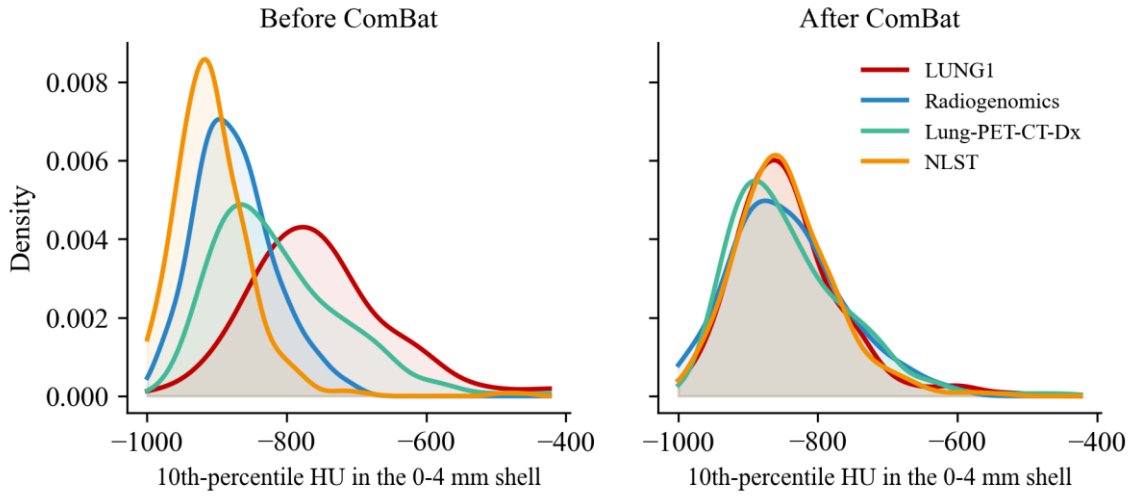


Figure 3: Effect of ComBat on a shell feature (10th-percentile HU in the 0-4 mm shell) across all 1041 patients. Left: each hospital has its own location and spread. Right: the distributions are aligned after ComBat. (Fitted on all patients for illustration purposes only.)

3.2.5 Feature Scaling, Ranking and Selection

Features with >20% missing values or with zero variance are removed in each training fold and the missing values are filled with the training median. After ComBat, each feature is z-scored with the training mean and standard deviation, $\tilde{x}_{ig} = (x_{ig}^* - \mu_g^{\text{tr}}) / s_g^{\text{tr}}$. Features are then ranked according to their ability to separate ADC from SCC in each hospital:

$$\rho_g = \frac{1}{H} \sum_{h=1}^H |\text{AUC}_h(\tilde{x}_g) - 0.5| \quad (17)$$

where AUC_h is computed with Eq. (1) on hospital h only. Ranking within hospitals avoids selecting features that mainly separate hospitals. Going down the ranked list, a feature is skipped if its absolute correlation with a feature already kept is greater than 0.9, $|r_{gk}| = |\frac{1}{n} \sum_i \tilde{x}_{ig} \tilde{x}_{ik}| > 0.9$. The first $k \in \{10, 25, 50\}$ kept features are used, and k is a hyperparameter.

3.2.6 Classifiers

Four classical models are trained on each block, all with the sample weights w_i of Eq. (4)

(implemented with scikit-learn [26] and LightGBM [27]).

(1) L2 logistic regression models the probability of ADC with the sigmoid function and minimises the weighted log-loss plus an L2 penalty:

$$p(x) = \frac{1}{1 + e^{-(\beta^T x + \beta_0)}} \quad (18)$$

$$\min_{\beta, \beta_0} \frac{1}{2} \|\beta\|_2^2 + C \sum_{i=1}^n w_i [-y_i \log p(x_i) - (1 - y_i) \log(1 - p(x_i))] \quad (19)$$

A small C means a strong penalty, so the weights stay small and the model stays simple.

(2) Support vector machine [28] with labels $t_i = 2y_i - 1 \in \{-1, +1\}$ finds the boundary with the largest possible margin, allowing some violations ξ_i :

$$\min_{v, b, \xi} \frac{1}{2} \|v\|^2 + C \sum_i w_i \xi_i \text{ s.t. } t_i(v^T \phi(x_i) + b) \geq 1 - \xi_i, \xi_i \geq 0 \quad (20)$$

Its decision function uses the radial basis function (RBF) kernel allowing for a curved boundary:

$$f(x) = \sum_i \alpha_i t_i K(x_i, x) + b, K(x, x') = \exp(-\gamma_K \|x - x'\|^2), \gamma_K = \frac{1}{p\text{Var}(X)} \quad (21)$$

(3) LightGBM [27] builds 200 small decision trees one after another. Each new tree h_m is fitted to the errors (negative gradients) of the current model and added with a learning rate $\eta = 0.03$, for $m = 1, \dots, 200$:

$$F_m(x) = F_{m-1}(x) + \eta h_m(x), r_{im} = y_i - \sigma(F_{m-1}(x_i)) \quad (22)$$

The trees have 4 or 8 leaves and at least 10 patients per leaf, and each tree uses 80% of the patients and 50% of the features. This limits overfitting.

(4) Ridge logistic regression on all features is Eq. (19) with all the features of the block (no top-k) and a very small $C \in [10^{-4}, 10^{-2}]$. It picks up a weak signal spread thinly over many correlated features.

3.2.7 Ensemble and Score Fusion

Each model outputs a real-valued score $f_m(x)$ (decision function or probability for LightGBM). The four scores are on different scales, and are each standardised to the mean μ_m and standard deviation σ_m of the training scores, and the four averaged into a block score:

$$s_b(x) = \frac{1}{4} \sum_{m=1}^4 \frac{f_m(x) - \mu_m}{\sigma_m} \quad (23)$$

The two block scores are z-scored and then averaged with equal, fixed weights, so nothing is fitted at this stage:

$$s(x) = \frac{1}{2} [z(s_{\text{shell}}(x)) + z(s_{\text{pos}}(x))], \hat{y} = \mathbb{1}[s(x) > 0] \quad (24)$$

A positive score means ADC. As the training weights balance the classes, the natural threshold is 0.

3.2.8 Evaluation Metrics

The main metric is the site-balanced AUC of Eq. (5). For the unseen-hospital and external tests, the ordinary AUC is used, since only one hospital is scored at a time. At the threshold of Eq. (24) we also report

$$\text{Se} = \frac{TP}{TP + FN}, \text{Sp} = \frac{TN}{TN + FP}, \text{BA} = \frac{\text{Se} + \text{Sp}}{2} \quad (25)$$

where ADC is the positive class. Uncertainty is quantified with a stratified bootstrap [29]: patients are resampled with replacement within each (hospital, class) group $B = 2000$ times, and the metric is recomputed each time. The 95% confidence interval is $[\hat{\theta}_{(2.5)}^*, \hat{\theta}_{(97.5)}^*]$. The hospital information remaining in the features is measured by a hospital detector, a one-vs-rest logistic regression trained to predict the hospital and scored on the test fold,

$$\text{Leak} = \frac{1}{H} \sum_{h=1}^H \text{AUC}(\mathbb{1}[h_i = h], g_h(\tilde{x}_i)) \quad (26)$$

A value close to 0.5 indicates that the hospital can no longer be identified. For visualisation we also use PCA which projects the features onto the two eigenvectors u_1, u_2 of the covariance matrix $\Sigma = \frac{1}{n} \tilde{X}^T \tilde{X}$ with the largest eigenvalues.

3.3 System Block Diagram

The entire pipeline from a raw CT scan to the ADC/SCC decision is illustrated in Figure 4. All steps are automatic and the same frozen pipeline was used for both external test sets. The steps are divided into four colours: image preparation, tumour and regions, features and model.

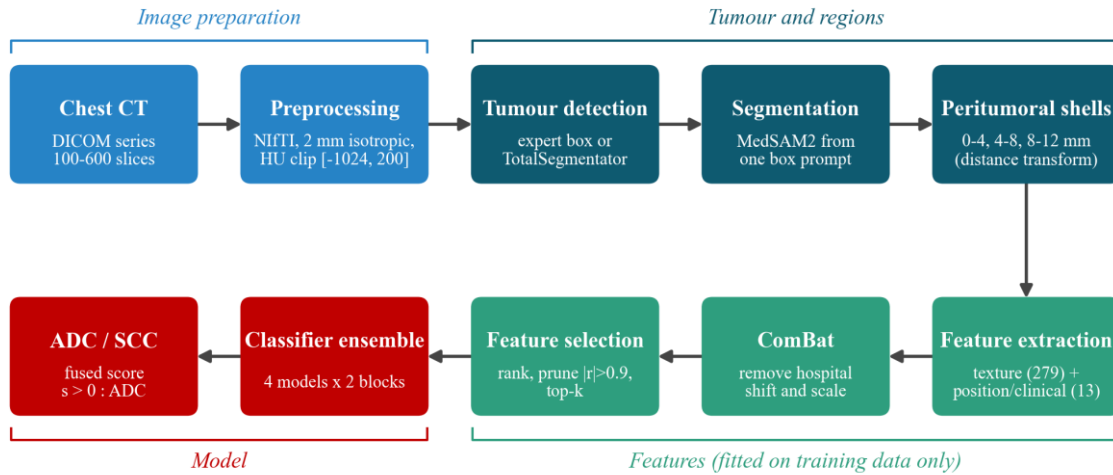


Figure 4: Block diagram of the complete pipeline. The steps in the lower row are fitted only on the training data and then frozen.

- Chest CT and preprocessing: DICOM to Nifti, 2 mm isotropic resampling, HU clipping (Eq. (7), Eq. (8)).
- Tumour detection and segmentation: expert box or TotalSegmentator box then MedSAM2 outline.
- Peritumoral shells: three distance bands surrounding the tumour (Eq. (10)).
- Feature extraction: 279 shell texture and 13 position/clinical features.
- ComBat, selection and classifiers: Eq. (13) to Eq. (22), inside each training fold.
- Decision: fused score s of Eq. (24); $s > 0$ means ADC.

3.4 Model Architecture

The detailed classifier part is shown in Figure 5. It has two parallel branches, one per feature block. Each branch has its own preparation chain and own four-model ensemble. The design is intentionally small: each model sees at most 50 features (or all 279 for the ridge model), so the whole classifier has

a few hundred parameters plus the trees, compared with millions in a CNN. Table 3 presents settings.

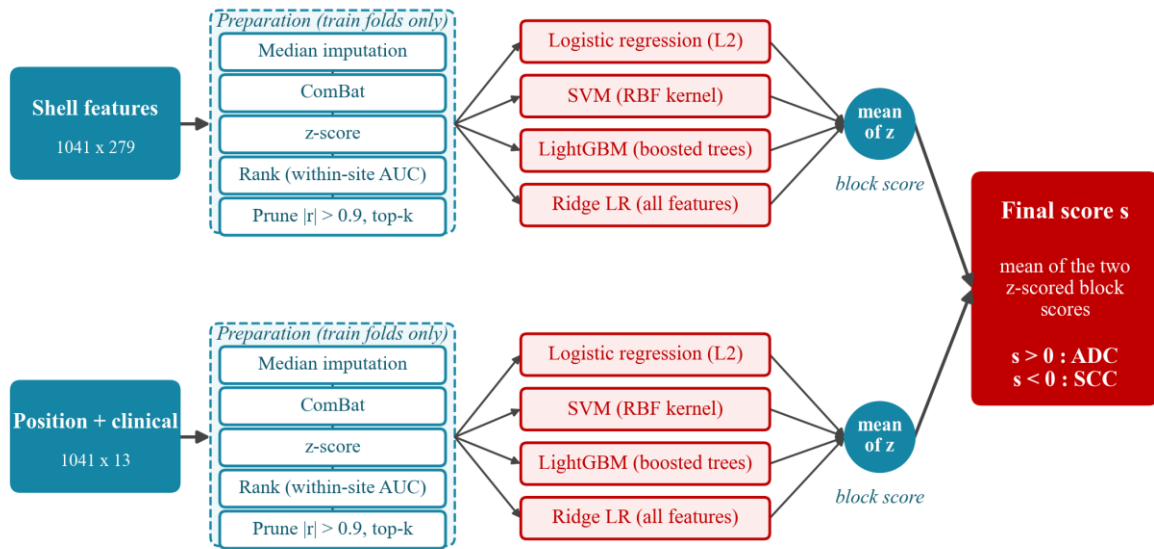


Figure 5: Architecture of the final classifier: two feature blocks, each containing a preparation chain and a four-model ensemble, fused by averaging z-scored block scores.

Component	Setting
Input per patient	279 shell features + 13 position/clinical features
Missing values	drop feature if > 20% missing; else training median
Harmonisation	ComBat with empirical Bayes, per fold; label-free mapping of new hospitals
Selection	within-hospital AUC ranking, correlation pruning at 0.9, top-k with k in {10, 25, 50}
Logistic regression	L2, C in {0.01, 0.1, 1}
SVM	RBF kernel, gamma = scale, C in {0.1, 1, 10}
LightGBM	200 trees, learning rate 0.03, leaves in {4, 8}, min. 10 per leaf, subsample 0.8, feature fraction 0.5
Ridge (all features)	L2 logistic regression, C in {0.0001, 0.0003, 0.001, 0.003, 0.01}
Sample weights	$1 / (H \times n \text{ of the hospital-class group})$, Eq. (4)
Block score	mean of the four training-standardised scores, Eq. (23)
Final score	mean of the two z-scored block scores; ADC if $s > 0$, Eq. (24)

Table 3: Settings of the final model

3.5 Training and Validation Design

Two complementary validation schemes were used (Figure 6).

Repeated nested cross-validation

Patients are divided into 5 folds stratified by hospital and class, so that each fold has all eight (hospital, class) groups. The model is trained on four folds and tested on the fifth. Inside the training part, an inner 3-fold cross-validation chooses the hyperparameters (k , C , number of leaves) by the site-balanced AUC, and the model is then refitted on the full training part. The outer loop is repeated for 5 random seeds (42-46), for a total of 25 test runs. Nested cross-validation avoids the optimistic bias introduced by tuning hyperparameters on the test fold [30].

Leave-one-hospital-out (LOSO)

The model is trained (with the same inner tuning) on three hospitals and tested on the remaining one, which is mapped into the training reference by label-free ComBat. This is repeated for all four

hospitals, which is the closest to simulating using the model on a new hospital.

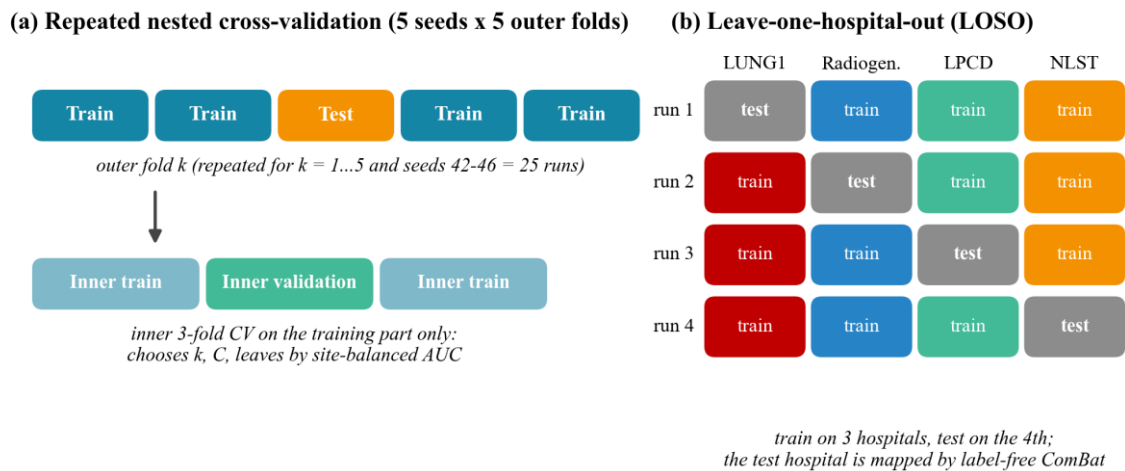


Figure 6: Validation design: (a) repeated nested cross-validation with inner 3-fold loop for hyperparameter tuning; (b) leave-one-hospital-out testing.

Model	Hyperparameters searched	Most often chosen (25 folds)
Logistic regression	C in {0.01, 0.1, 1}; k in {10, 25, 50}	C = 0.01 (strongest penalty)
SVM (RBF)	C in {0.1, 1, 10}; k in {10, 25, 50}	C = 0.1
LightGBM	leaves in {4, 8}; k in {10, 25, 50}	4 leaves
Ridge (all)	C in {0.0001, 0.0003, 0.001, 0.003, 0.01}	C = 0.003 (shells), 0.01 (position)

Table 4: Hyperparameter grid; the selection criterion is the site balanced AUC of the inner folds

3.6 Locked External Test Protocol

A model that is tested, adjusted and tested again slowly turns its test set into a training set. To avoid this, the two external datasets were handled according to the protocol in Figure 7. Before any external image was processed, a pre-registration document fixed the model (shells + position, frozen after Week 13), the exclusion rules and the primary metric. The images were processed by the frozen automatic pipeline. The model was fitted once on all 1041 development patients, and the new dataset was mapped as one new site with label-free ComBat. The prediction file was saved together with its SHA-256 hash, a 64-character fingerprint that changes completely if even one byte of the file changes. Only then were the labels opened, once, by a scoring script that refuses to run a second time.

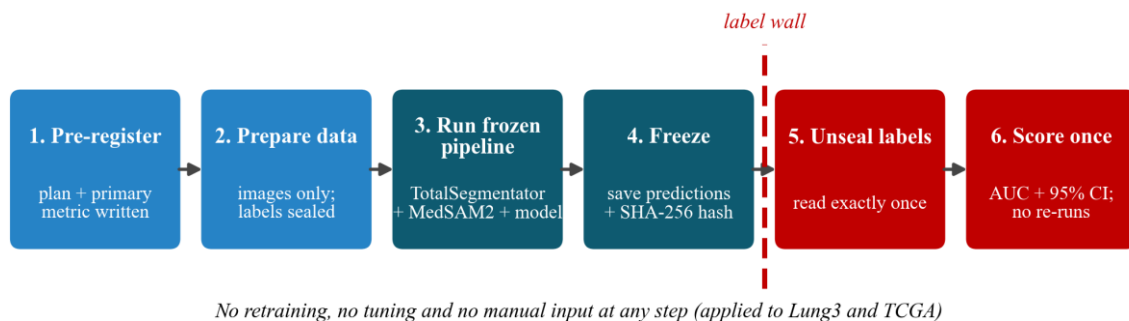


Figure 7: Locked protocol for external tests. The labels are only revealed after the predictions have been frozen and hashed.

3.7 Core Source Code

The project was written in Python and the full code is available in the GitHub repository provided in section 7.3. The most important parts are the site-balanced metric and feature ranking (Table 5), the ComBat harmoniser (Table 6) and nested model selection, ensembling and leave-one-hospital-out testing (Table 7).

<pre> def sb_weights(site, y): """1 / n_{site,label}, mean 1: every site counts equally, each site 50/50.""" key = pd.Series(site).astype(str) + " " + pd.Series(y).astype(str) cnt = key.map(key.value_counts()) w = 1.0 / cnt.values return w * len(w) / w.sum() def sb_auc(y, s, site): return float(roc_auc_score(y, s, sample_weight=sb_weights(site, y))) def univariate_score(X, y, site): """Mean over sites of the within-site AUC, folded around 0.5 (vectorised).""" sc = np.zeros(X.shape[1]) sites = np.unique(site) for g in sites: m = site == g yy = y[m] n1, n0 = yy.sum(), (1 - yy).sum() r = rankdata(X[m], axis=0) auc = (r[yy == 1].sum(0) - n1 * (n1 + 1) / 2.0) / (n1 * n0) sc += np.abs(auc - 0.5) return sc / len(sites) def site_leak_auc(Xtr, Xte, site_tr, site_te, y_tr, y_te): """mean one-vs-rest AUC of a hospital classifier (label-balanced weights).""" out = [] for s in np.unique(site_tr): t_tr, t_te = (site_tr == s).astype(int), (site_te == s).astype(int) if t_te.min() == t_te.max(): continue m = LogisticRegression(C=0.1, max_iter=5000).fit(Xtr, t_tr, sample_weight=sb_weights(site_tr, y_tr)) out.append(roc_auc_score(t_te, m.decision_function(Xte), sample_weight=sb_weights(site_te, y_te))) return float(np.mean(out)) </pre>	<pre> def select(self, k): if k == 0: # every feature (dense ridge model) return list(range(self.Xtr_.shape[1])) thr = self.cfg["features"]["correlation_prune"] if not hasattr(self, "_sel_cache"): self._sel_cache = {} kmax = max(self.cfg["features"]["top_k_grid"]) if "full" not in self._sel_cache: kept = [] C = None for j in self.order_: if len(kept) >= kmax: break if kept: v = self.Xtr_[j] r = np.abs(C.T @ v) / len(v) if r.max() > thr: continue kept.append(j) C = self.Xtr_[j, kept] self._sel_cache["full"] = kept return self._sel_cache["full"][:k] def fit_model(cfg, kind, hp, X, y, w, seed): if kind in ("logreg", "ridge_all"): mdl = LogisticRegression(C=hp["C"], max_iter=5000) mdl.fit(X, y, sample_weight=w) return mdl if kind == "svm": mdl = SVC(C=hp["C"], kernel="rbf", gamma=cfg["models"]["svm"]["gamma"]) mdl.fit(X, y, sample_weight=w) return mdl import lightgbm as lgb g = cfg["models"]["lgbm"] mdl = lgb.LGBMClassifier(n_estimators=g["n_estimators"], learning_rate=g["learning_rate"], num_leaves=hp["num_leaves"], min_child_samples=g["min_child_samples"], subsample=g["subsample"], subsample_freq=1, colsample_bytree=g["colsample_bytree"], random_state=seed, n_jobs=1, verbose=-1) mdl.fit(X, y, sample_weight=w) return mdl def score(mdl, X): if hasattr(mdl, "decision_function"): return mdl.decision_function(X) return mdl.predict_proba(X)[: , 1] </pre>
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Table 5: Source code: site-balanced weights and AUC, within-hospital feature ranking, hospital detector (left); correlation-pruned top-k selection and the four classifiers (right)

<pre> def _eb_prior(gamma_hat, delta_hat): g_bar, t2 = gamma_hat.mean(), gamma_hat.var(ddof=1) m, v = delta_hat.mean(), delta_hat.var(ddof=1) v = max(v, 1e-12) a = (2.0 * v + m * m) / v b = (m * v + m ** 3) / v return g_bar, max(t2, 1e-12), a, b def _eb_shrink(gamma_hat, delta_hat, n, iters=100, tol=1e-4): g_bar, t2, a, b = _eb_prior(gamma_hat, delta_hat) g, d = gamma_hat.copy(), delta_hat.copy() for _ in range(iters): g_new = (n * t2 * gamma_hat + d * g_bar) / (n * t2 + d) ss = n * (delta_hat + (gamma_hat - g_new) ** 2) d_new = (0.5 * ss + b) / (n / 2.0 + a - 1.0) conv = max(np.max(np.abs(g_new - g) / (np.abs(g) + 1e-8)), np.max(np.abs(d_new - d) / (np.abs(d) + 1e-8))) g, d = g_new, d_new if conv < tol: break return g, d </pre>	<pre> class Harmoniser: def __init__(self, method="none", empirical_bayes=True): self.method = method self.eb = empirical_bayes def fit(self, X, site, y): X = np.asarray(X, float) site = np.asarray(site) self.sites_ = list(np.unique(site)) if self.method == "none": return self bal = self.method == "balanced_combat" loc, var, n = {}, {}, {} for s in self.sites_: m = site == s loc[s], var[s] = _site_stats(X[m], None if y is None else np.asarray(y)[m], bal) n[s] = int(m.sum()) if bal: # every site counts equally in the reference self.alpha_ = np.mean([loc[s] for s in self.sites_], axis=0) pooled = np.mean([var[s] for s in self.sites_], axis=0) else: tot = sum(n.values()) self.alpha_ = sum(n[s] * loc[s] for s in self.sites_) / tot pooled = sum(n[s] * var[s] for s in self.sites_) / tot self.sigma_ = np.sqrt(np.maximum(pooled, 1e- 12)) self.gamma_, self.delta_ = {}, {} for s in self.sites_: gh = (loc[s] - self.alpha_) / self.sigma_ dh = np.maximum(var[s] / self.sigma_ ** 2, 1e-8) if self.eb and len(gh) > 2: gh, dh = _eb_shrink(gh, dh, n[s]) self.gamma_[s], self.delta_[s] = gh, dh return self def transform(self, X, site): X = np.asarray(X, float) if self.method == "none": return X.copy() site = np.asarray(site) out = np.empty_like(X) for s in np.unique(site): if s not in self.gamma_: raise KeyError("site %r unseen at fit time; use fit_new_site" % s) m = site == s z = (X[m] - self.alpha_) / self.sigma_ out[m] = self.sigma_ * (z - self.gamma_[s]) / np.sqrt(self.delta_[s]) + self.alpha_ return out def fit_new_site(self, X_new, name, balanced, w1=None): X_new = np.asarray(X_new, float) loc, var = _site_stats(X_new, None, balanced, w1) self.gamma_[name] = (loc - self.alpha_) / self.sigma_ self.delta_[name] = np.maximum(var / self.sigma_ ** 2, 1e-8) return self </pre>
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Table 6: Source code: empirical-Bayes priors and shrinkage (left); ComBat harmoniser with label-free mapping of a new hospital (right)

<pre> def select_and_fit(cfg, X, y, site, seed, harm): """Inner-CV hyper-parameter choice on (X, y, site), then refit on all of it.""" strata = np.array([f"{a} {b}" for a, b in zip(site, y)]) inner = StratifiedKFold(cfg["experiment"]["n_inner_folds"], shuffle=True, random_state=seed) sc = {k: {i: [] for i in range(len(grid(cfg, k)))} for k in MODELS} for a, b in inner.split(X, strata): pr = Prep(harm, cfg).fit(X[a], y[a], site[a]) Xa, Xb = pr.Xtr_, pr.transform(X[b], site[b]) w = sb_weights(site[a], y[a]) for k in MODELS: for i, hp in enumerate(grid(cfg, k)): sel = pr.select(hp["k"]) m = fit_model(cfg, k, hp, Xa[:, sel], y[a], w, seed) sc[k][i].append(sb_auc(y[b], score(m, Xb[:, sel]), site[b])) pr = Prep(harm, cfg).fit(X, y, site) w = sb_weights(site, y) fitted = {} for k in MODELS: best = max(sc[k], key=lambda i: np.mean(sc[k][i])) hp = grid(cfg, k)[best] sel = pr.select(hp["k"]) m = fit_model(cfg, k, hp, pr.Xtr[:, sel], y, w, seed) s = score(m, pr.Xtr[:, sel]) fitted[k] = (m, sel, s.mean(), s.std() + 1e-8, dict(hp, inner=float(np.mean(sc[k][best])))) return pr, fitted def ens(fitted, Z): return np.mean([(score(m, Z[:, sel]) - mu) / sd for m, sel, mu, sd, _ in fitted.values()], axis=0) </pre>	<pre> def loso_one(cfg, X, y, site, held, seed, harm): tr, te = site != held, site == held pr, fitted = select_and_fit(cfg, X[tr], y[tr], site[tr], seed, harm) # label-free mapping of the held-out hospital into the training reference Xte_raw = X[te][:, pr.keep_] Xte_raw = np.where(np.isnan(Xte_raw), pr.med_, Xte_raw)[: , pr.keep2_] if harm != "none": pr.h_.fit_new_site(Xte_raw, held, balanced=False) Zte = (pr.h_.transform(Xte_raw, np.array([held] * te.sum())) - pr.mu_) / pr.sd_ s = ens(fitted, Zte) return pd.DataFrame({"idx": np.where(te)[0], "seed": seed, "held_out": held, "score": s}) </pre>
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Table 7: Source code: inner-CV hyperparameter selection and ensemble (left); leave-one-hospital-out test with label-free harmonisation (right)

4 Implementation

4.1 Description

Environment

All work was done on one laptop. The hardware and main software are listed in Table 8. MedSAM2, TotalSegmentator and the main pipeline require different versions of PyTorch, so separate Python environments were necessary.

Item	Details
Hardware	Laptop, NVIDIA RTX 3050 GPU (4 GB), 16 GB RAM, Windows 11
Language	Python 3.9 (main pipeline), 3.12 (MedSAM2)
Image I/O and resampling	SimpleITK, pydicom, dcm2niix-style conversion, NumPy, SciPy
Features	PyRadiomics 3 (texture), own code for position features and shells
Machine learning	scikit-learn 1.5.2, LightGBM 4.6.0, PyTorch 2.6 (deep baselines), MONAI 1.3.2
Segmentation	MedSAM2 (box-prompted), TotalSegmentator 2.18 (lung nodules)
Data download	NBIA Data Retriever (TCIA), s5cmd and idc-index (IDC)
Reporting	Matplotlib, python-pptx, python-docx

Table 8: Implementation environment

Data preparation

The DICOM series were downloaded (approx. 90 GB total), checked for geometry (spacing, orientation, missing slices) and converted to NIfTI. Figure 8 shows a LUNG1 scan before and after preprocessing. In the original grid the coronal view is squashed as the slices are 3mm apart while

pixels are roughly 1mm. After resampling the anatomy looks correct in proportion and the HU window allows easy visualisation of lung and soft tissue.

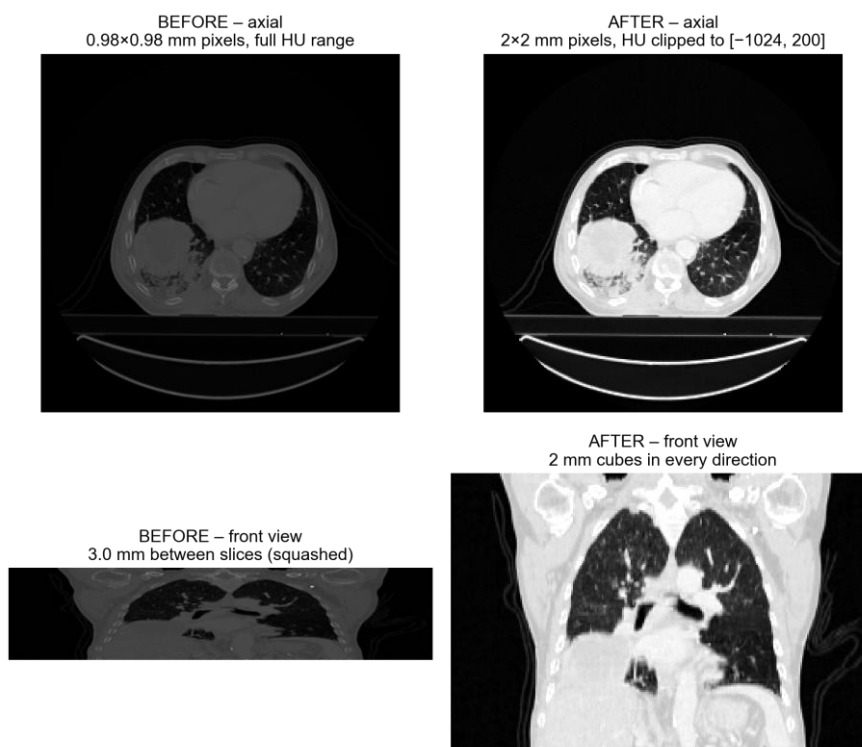


Figure 8: Preprocessing of patient LUNG1-001. Original axial and coronal views (left) and the same views after 2 mm isotropic resampling and HU clipping (right).

Segmentation

Our first segmentation method kept voxels above -750 HU inside the box and cleaned the result with morphology. It worked with tight boxes but leaked into the chest wall when the box was loose, which was the case for the NLST boxes. MedSAM2 was therefore used for all 1041 patients (38 minutes on the laptop GPU, no failures). The settings were chosen on 40 expert outlines using the worst-case Dice, without looking at any ADC/SCC label: lung window [-1350, 150] HU, full 512 x 512 slices, one box prompt on the slice with the largest box, propagation in both directions, and the result limited to the box enlarged by 3 mm. An example is shown in Figure 9 and the validation on all 343 expert outlines in Figure 10.

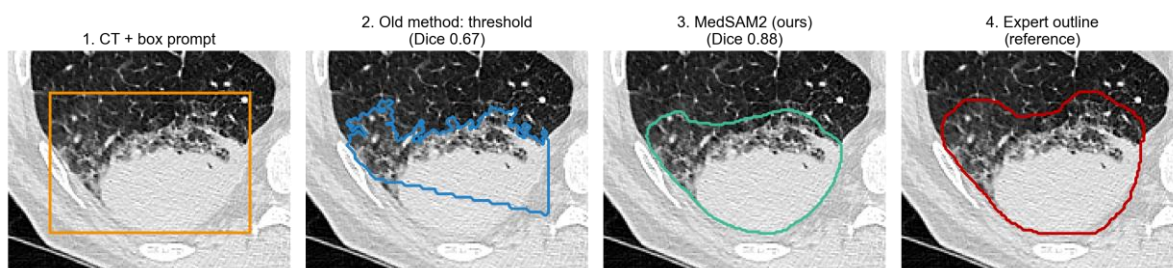


Figure 9: Segmentation of a single Radiogenomics patient: box prompt, old threshold method (Dice 0.67), MedSAM2 (Dice 0.88) and expert outline.

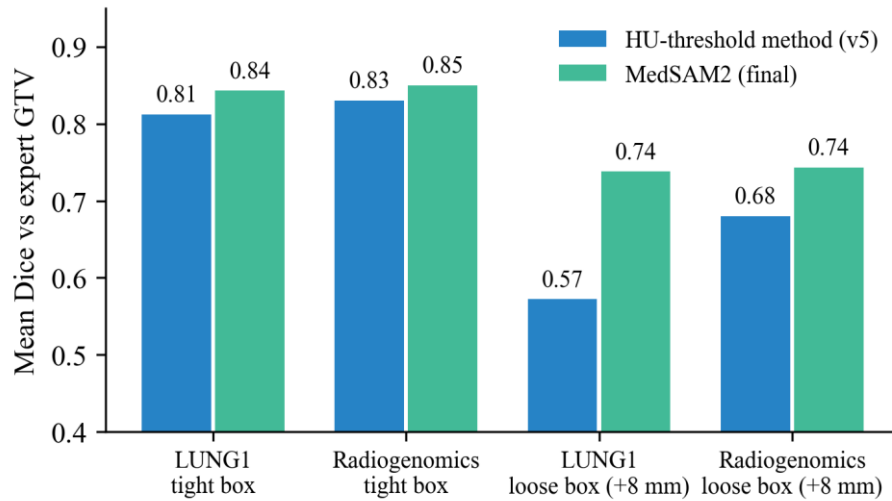


Figure 10: Mean Dice against expert outlines for tight boxes and loose boxes (+8 mm on each side, mimicking an automatic detector). MedSAM2 is better in all four cases and much better with loose boxes.

Feature extraction

The shells were built with a Euclidean distance transform (Eq. (10)); Figure 11 depicts them on a real scan. The PyRadiomics extraction for all regions took ~20 minutes and position features ~13 minutes using parallel jobs with no failures in the development data. Since extraction was run once and the features saved, model training never re-calculated them.

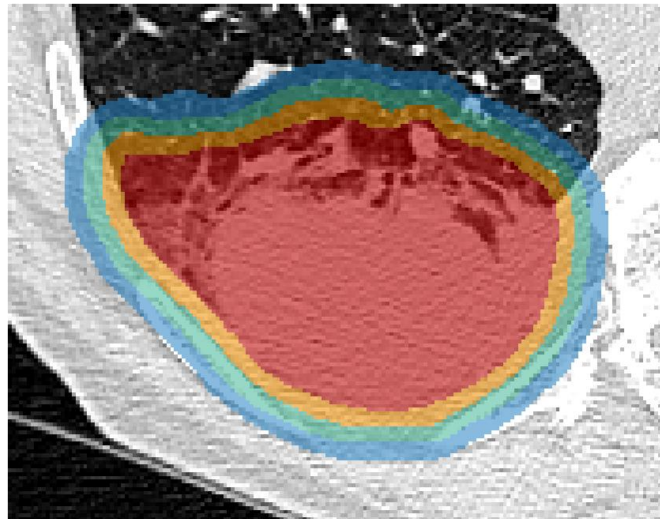


Figure 11: Tumour (GTV) and the three peritumoral shells at 0-4, 4-8 and 8-12 mm on an axial CT slice.

Training

The models are small so a full 5 x 5 nested cross-validation of one feature block takes about 10 seconds to a few minutes with 14 parallel jobs. All splits, random seeds, chosen hyperparameters, out-of-fold predictions, metrics and package versions are saved for reproducibility.

4.2 Implementation Challenges

- Different DICOM types. Eight Lung-PET-CT-Dx patients had annotations only on a secondary-capture image without 3-D geometry, so they were excluded by rule.
- NLST with loose boxes. The threshold segmentation included the chest wall, as the NLST boxes were approximately three times the size of the tumour. MedSAM2 solved this.
- Software conflicts. During one run, installing a foundation-model package replaced the GPU

version of PyTorch with a CPU version. The environment was repaired, and later packages were installed without changing the existing ones.

- Large downloads on Windows. The `s5cmd` tool could not rename files across drives, so downloads were run from the destination drive with relative paths.
- Limited GPU memory (4 GB). Fine-tuning used a frozen backbone with only the last block trained, together with cached augmented crops.

5 Design Analysis and Evaluation

5.1 Novelty

We did not invent ComBat, MedSAM2 or weighted AUC individually. The novelty of the project lies in the way they are combined and in the way the model is evaluated:

- Direct measurement of the hospital shortcut for ADC/SCC. We show that the hospital name alone yields a pooled AUC of 0.71 and use a site-balanced AUC that provably gives 0.5 to any hospital-only predictor (Eq. (6)). The honest score increased from 0.642 (naive model) to 0.722.
- Fully automatic, uniform tumour outlining. TotalSegmentator and a single MedSAM2 box prompt give masks at Dice 0.85 against experts, using the same method in every hospital, so that the source of the mask is not a hidden fingerprint.
- Four hospitals, three countries, 1,041 patients. To the best of our knowledge this is one of the largest public multi-hospital datasets for this task and enabled leave-one-hospital-out testing (0.714).
- Locked, pre-registered external tests. The plan was written first, predictions were hashed before the labels were opened, and each test was scored once (Lung3 0.686, TCGA 0.585).
- A fair comparison between simple and deep models. Attention-MIL, CNN features, two foundation models and adversarial fine-tuning were all tested under the same protocol, and all of them scored lower.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

The model is a decision support tool. It gives a score, not a diagnosis, and a biopsy is still needed to decide treatment and for gene testing. At the default threshold (Section 5.3.3), sensitivity is 0.61 and specificity is 0.68, so wrong predictions are frequent enough that the score should never be used alone. This leads to a number of safety choices: the threshold is fixed in advance, the model reports results per hospital, and we state clearly that it should not be used on 5 mm CT, where external performance was at chance. The pipeline adds no radiation dose and no procedural risk because it uses only the CT the patient already has.

5.2.2 Considerations to environment

We used small classical models instead of large networks. Training the final model takes a few minutes on a laptop GPU, and the whole project used around 10-15 kWh of electricity (Section 8). By using public data rather than taking new scans, we avoided additional scanning and the energy and radiation

cost that comes with it. The downloaded data were stored on a single disk and processed once, and the extracted features were saved so that experiments did not repeat expensive computation.

5.2.3 Considerations to cultural and societal needs

CT scanners are more easily available in Bangladesh and similar countries, while biopsy facilities and pathologists are mostly confined to the big hospitals. A CT-based tool could help prioritise patients for biopsy or guide the choice of sample. High smoking rates among men contribute to the share of SCC. Our data are from the Netherlands, the USA and China so the performance in a South Asian population is unknown and needs testing locally prior to any use. The site-balanced evaluation is also a societal concern: it means that a hospital with fewer patients is not ignored by the metric.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

In each experiment one part of the system was changed while the folds, seeds and metric were kept the same, so that differences could be attributed to that part. The experiments are listed in Table 9.

#	Question	What was varied	Metric
E1	How strong is the hospital shortcut?	hospital-only predictor vs naive vs final model	pooled and site-balanced AUC
E2	Does attention over local patches help?	mean vs gated-attention MIL, patch features, ROI	AUC (LUNG1)
E3	Which feature block carries signal?	7 blocks alone and 127 fusions	site-balanced AUC, LOSO
E4	Does harmonisation remove the fingerprint?	ComBat on/off	hospital-detector AUC, AUC
E5	Do better masks help?	threshold masks vs MedSAM2	Dice, AUC
E6	Do more hospitals help?	2, 3 and 4 training hospitals	LOSO AUC
E7	Do deep features beat radiomics?	FMCIB, CT-FM, fine-tuned CNN with adversary	AUC, hospital-detector AUC
E8	Does the frozen model transfer?	locked Lung3 and TCGA	AUC with 95% CI

Table 9: Design of experiments

5.3.2 Data Collection

All data are de-identified and public. Table 10 gives a summary of the six cohorts. The development set consists of 1041 patients (697 ADC, 344 SCC) from four sources [3, 31, 32, 33]. The NLST tumour boxes come from the Sybil project [34]. The Lung3 [35] and TCGA [36] datasets were kept as external test sets and were not opened until the end.

Cohort	Country / setting	Patients (ADC/SCC)	Typical slice	Tumour location	Role
LUNG1 (NSCLC-Radiomics)	Netherlands, radiotherapy, mostly stage III	202 (51/151)	3 mm	expert outline	development
NSCLC-Radiogenomics	USA, surgical	141 (112/29)	1.25 mm	expert outline	development
Lung-PET-CT-Dx	China, diagnostic	301 (244/57)	mostly 5 mm	expert boxes	development
NLST	USA, screening trial	397 (290/107)	2 mm	expert boxes (Sybil)	development
Lung3 (Radiomics-Genomics)	Netherlands, surgical	79 (44/35)	4-5 mm	automatic	locked test
TCGA-LUAD / LUSC	USA, 9 hospitals	84 (48/36)	5 mm (2/3 of scans)	automatic	locked test

Table 10: Datasets used in this project

Inclusion rules

Patients were required to have a pre-treatment CT series and a pathology label of ADC or SCC. Mixed or unclear types were excluded: in NLST, 155 patients whose first cancer was not ADC or SCC and 29 with more than one lung primary; in Lung-PET-CT-Dx, 8 patients with secondary-capture images; in Lung3, 10 patients with NSCLC not otherwise specified, large cell neuroendocrine or mixed histology; in TCGA, 19 patients without a usable CT and 1 with no detected nodule. Patients were selected using only metadata, never image-based predictions.

5.3.3 Results and Analysis

Early approach: attention-MIL on local patches

The project began with a weakly supervised attention-MIL model on LUNG1 [19, 20], where each patient was a bag of 1 cm radiomic patches. These trials were presented at the progress presentation and are summarised in Table 11. Attention did not reliably outperform simple mean pooling, and the best result after adding a second cohort (0.675) was later found to be partly inflated by the hospital shortcut, as a hospital detector reached an AUC of 0.998 on these two cohorts. This led to the redesigned system described in this report.

Trial	Setup	Result (AUC)	Finding
1	Gradient ROI, 29,184 patches, 74 features	0.604 / 0.568	attention unstable, lost to mean pooling
2	GTV + 8 mm rim, 37,464 patches	0.570 mean, 0.577 attention	wider region did not clearly help
3	mRMR [37] / LASSO [38] selection inside CV	0.593	small gain, CI included zero
4	Frozen ResNet-18 [39] patch embeddings	0.549 mean, 0.615 attention	first clear attention gain
5	CLAM-style instance loss	0.615 to 0.568	sharper attention, lower AUC
6	Global GTV and rim radiomics, late fusion	0.567 / 0.537	global models also weak on LUNG1
7	LUNG1 + Radiogenomics merged	0.675 (pooled)	partly hospital shortcut

Table 11: Summary of the first attention-MIL phase (prior to the progress presentation)

The hospital shortcut (E1)

Just the name of the hospital yields a pooled AUC of 0.710 and the naive radiomics model 0.727 (see Figure 2). Under the site-balanced AUC they are 0.497 and 0.642. All results below are for the site balanced metric.

Main cross-validation result

The final model achieved a site-balanced AUC of 0.722 (95% CI 0.687-0.762) over 5 seeds x 5 folds on 1041 patients, with a standard deviation over seeds of less than 0.01. The ROC curve inside each hospital (Figure 12) was built from the out-of-fold predictions averaged over the seeds. All the curves are above the diagonal. Lung-PET-CT-Dx is the easiest hospital (0.85) and LUNG1 the hardest (0.63). In LUNG1, 78% of patients are stage III, and at that stage ADC and SCC tumours are similar in size (median about 44 versus 48 ml), so size and position carry little information.

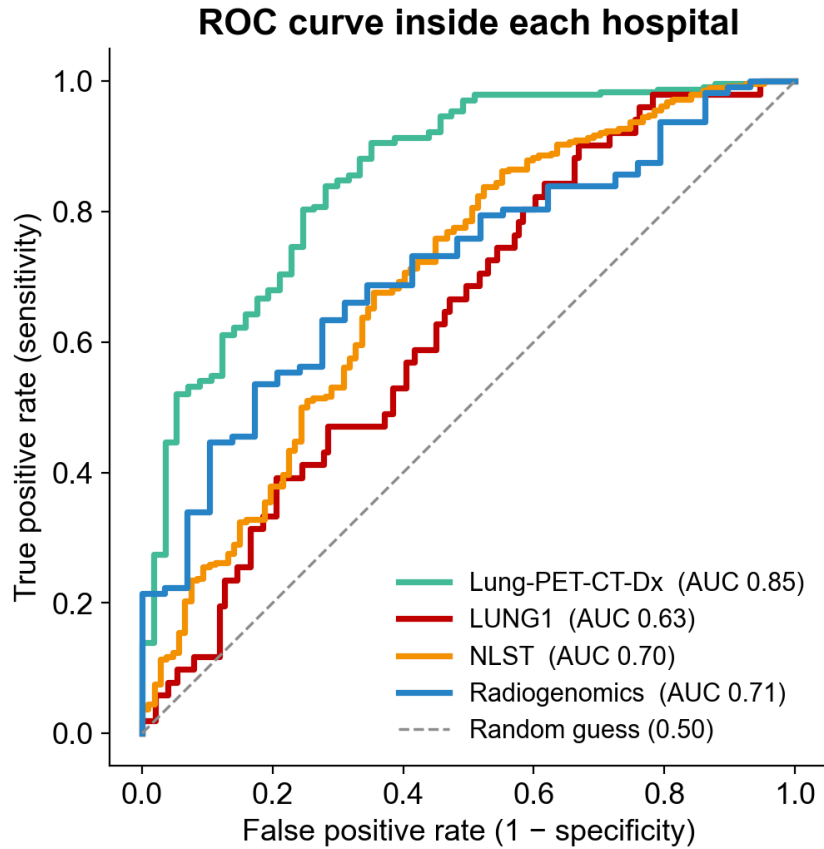


Figure 12: ROC curves of the final model within each hospital (out-of-fold predictions, 5 x 5 cross-validation). The dashed line is random guessing.

Figure 13 shows the distribution of the fused score by hospital and class. The ADC box is above the SCC box in each hospital, but the overlap is large, which corresponds to an AUC of around 0.7. The results at the fixed threshold $s = 0$ are shown in Table 12 and Figure 14.

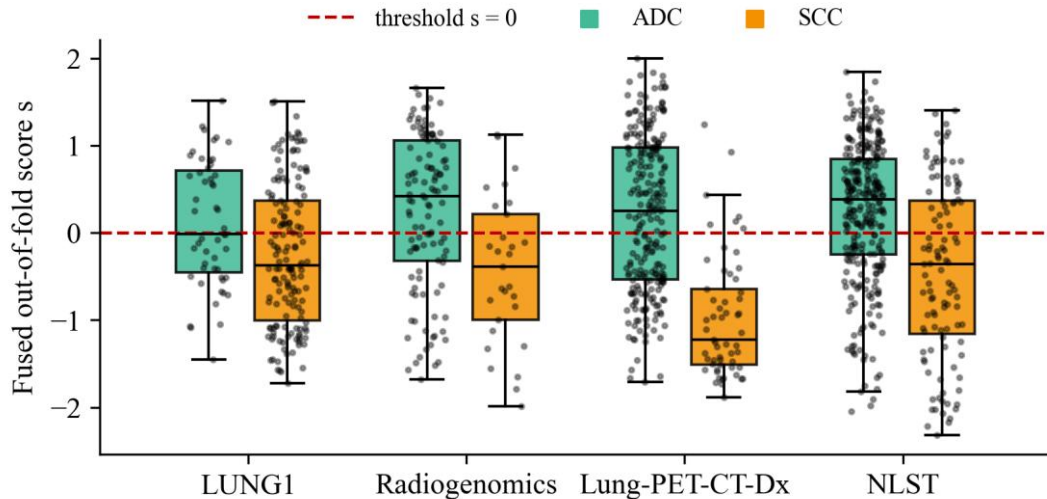


Figure 13: Fused out-of-fold score by hospital and class. Points are patients; the red line is the decision threshold.

Hospital	Patients (ADC/SCC)	AUC	Sensitivity	Specificity	Balanced acc.
LUNG1	202 (51/151)	0.631	0.490	0.623	0.556
Radiogenomics	141 (112/29)	0.710	0.616	0.724	0.670
Lung-PET-CT-Dx	301 (244/57)	0.852	0.561	0.877	0.719

Hospital	Patients (ADC/SCC)	AUC	Sensitivity	Specificity	Balanced acc.
NLST	397 (290/107)	0.694	0.662	0.645	0.653
All (site-balanced)	1041 (697/344)	0.722	0.607	0.680	0.644

Table 12: Cross-validation results per hospital at the threshold $s = 0$

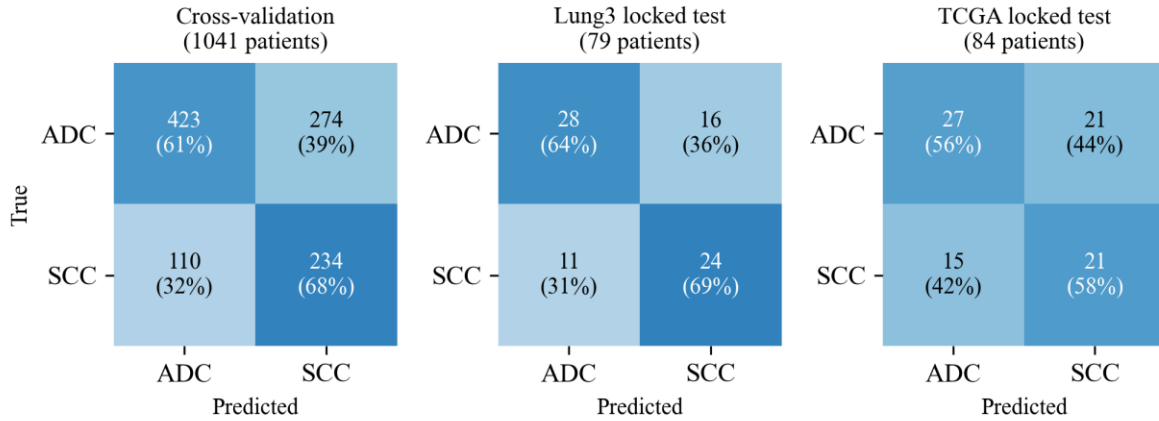


Figure 14: Confusion matrices at the threshold $s = 0$ for cross-validation and the two locked external tests. Percentages are given per true class.

Unseen hospitals (LOSO, E6)

The score for each hospital when it was included in the training data (pooled CV) versus left out completely is compared in Figure 15. The drop is very small (maximum 0.017) and the LOSO mean is 0.714, almost equal to the pooled 0.722. It implies that the model does not rely on memorising hospital style. More hospitals helped. With three training hospitals the LOSO mean was 0.678 and adding NLST raised it to 0.704 with the earlier masks, mostly by improving the held-out Lung-PET-CT-Dx score from 0.735 to 0.839.

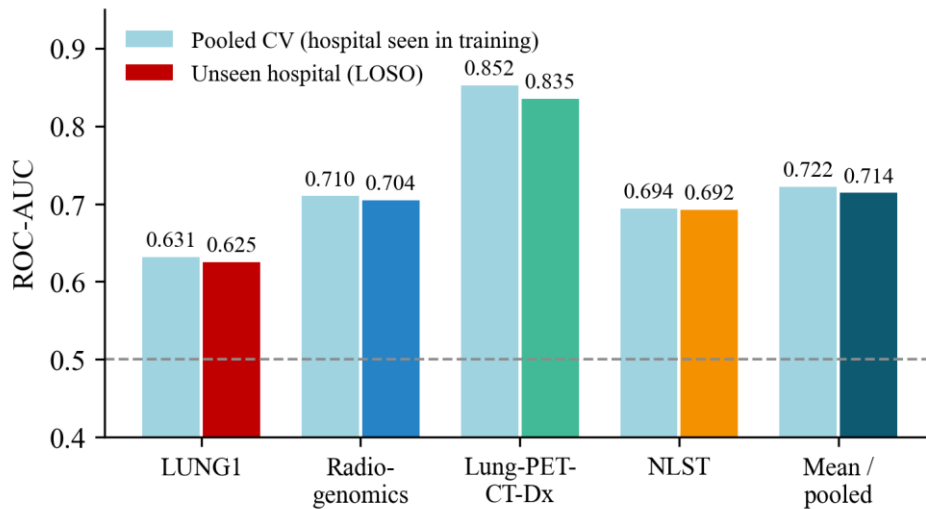


Figure 15: AUC of each hospital in pooled cross-validation (light bars) and when the hospital was left out of training (coloured bars).

Harmonisation (E4)

Figure 16 shows the shell features projected by PCA. Before ComBat, LUNG1 forms its own cloud; after ComBat the hospitals overlap. Figure 17 measures this with the hospital detector of Eq. (26): it decreased from 0.81-0.91 to 0.56-0.64 for the radiomics and position blocks. The site-balanced AUC changed by only -0.004 to +0.020 per block. So ComBat does not increase the score; it makes the score more reliable, because the remaining performance cannot be due to scanner style.

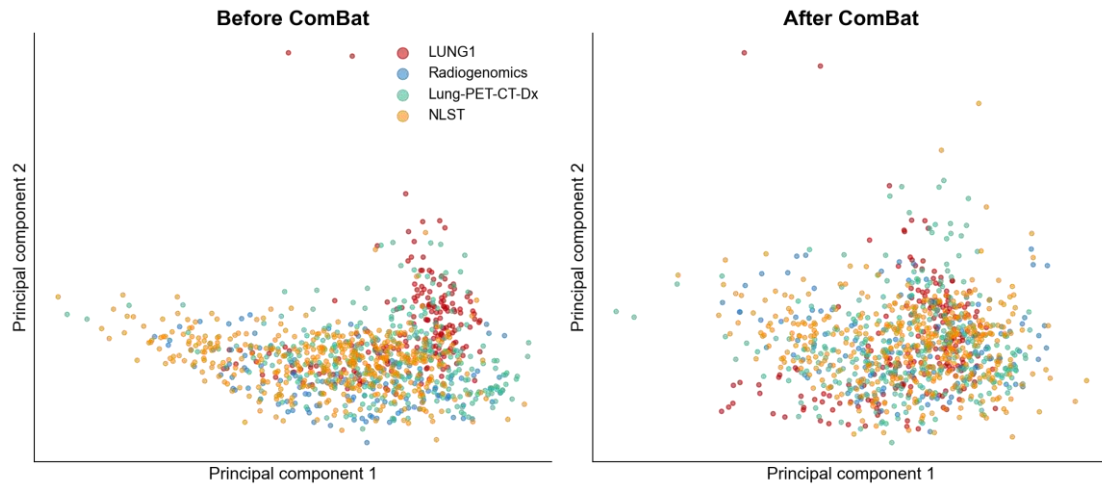


Figure 16: PCA of the 279 shell features before and after ComBat; each point is a patient, coloured by hospital.

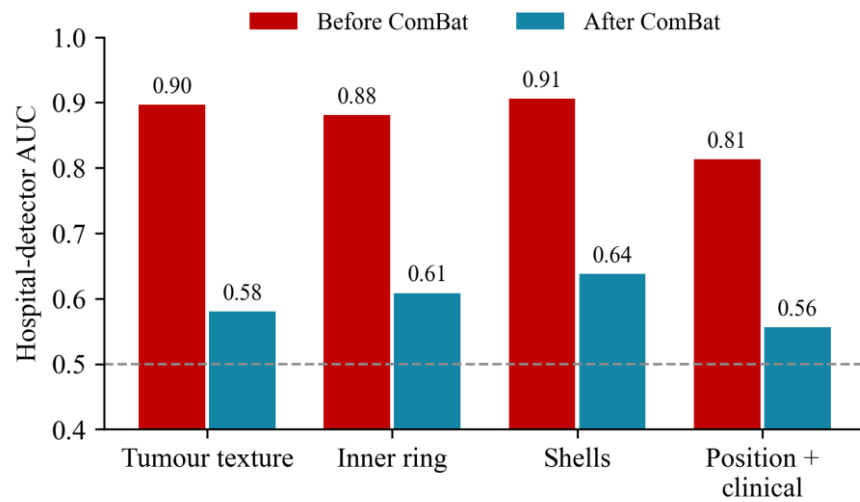


Figure 17: Hospital-detector AUC before and after ComBat (three-hospital stage). 0.5 means the hospital cannot be recognised.

Ablation (E3, E5)

The site-balanced AUC of each feature block and of the fusions, all with MedSAM2 masks and ComBat, is shown in Figure 18. The handcrafted blocks are close to each other (0.698-0.709), the two deep-feature blocks are clearly lower, and fusing shells with position gives the best pre-specified result. Table 13 lists the effect of each design change.

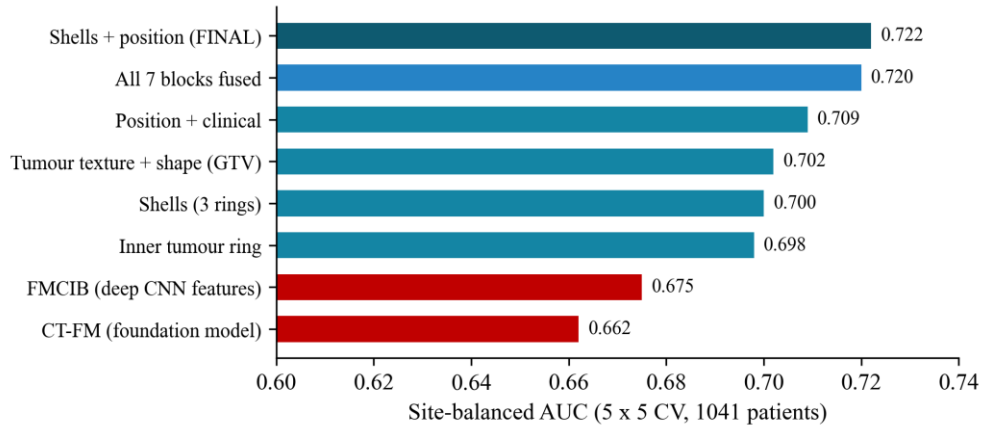


Figure 18: Ablation: site-balanced AUC of single feature blocks and fusions (5 x 5 CV, 1041 patients). Deep features are shown as red bars. Note that the x-axis begins at 0.6.

Change	Before	After	Conclusion
Threshold masks to MedSAM2 masks	0.714	0.722	better masks give better features
3 to 4 training hospitals (LOSO mean)	0.682	0.704	more diverse data helps transfer
Best single block to shells + position	0.709	0.722	the two blocks are complementary
Shells + position to all 7 blocks	0.722	0.720	more features add noise, not signal
ComBat off to on (hospital detector)	0.81-0.91	0.56-0.64	fingerprint largely removed
Single model to 4-model ensemble (shells)	0.677-0.697	0.700	ensemble is more stable

Table 13: Effect of individual design changes (site-balanced AUC unless otherwise stated)

Ensemble and tuning

Figure 19 compares each classifier with the ensemble. No single model is best in both blocks (ridge is best for shells, SVM for position), but the four-model ensemble is better than each individual model in both blocks, and fusing the two blocks adds a further 0.013. Figure 20 shows the hyperparameters selected by the inner cross-validation in the 25 outer folds. The simplest option was chosen almost every time: $C = 0.01$ for logistic regression, $C = 0.1$ for the SVM and 4-leaf trees. This suggests that the data are noisy and small, and that more flexible models would mostly fit noise.

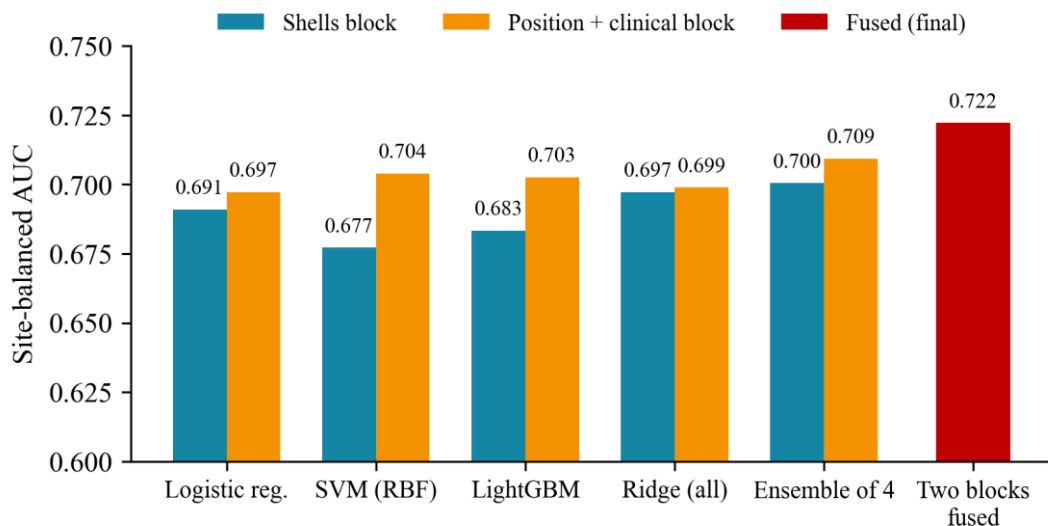


Figure 19: The site-balanced AUC of each classifier, the four-model ensemble in each block, and the fused final model.

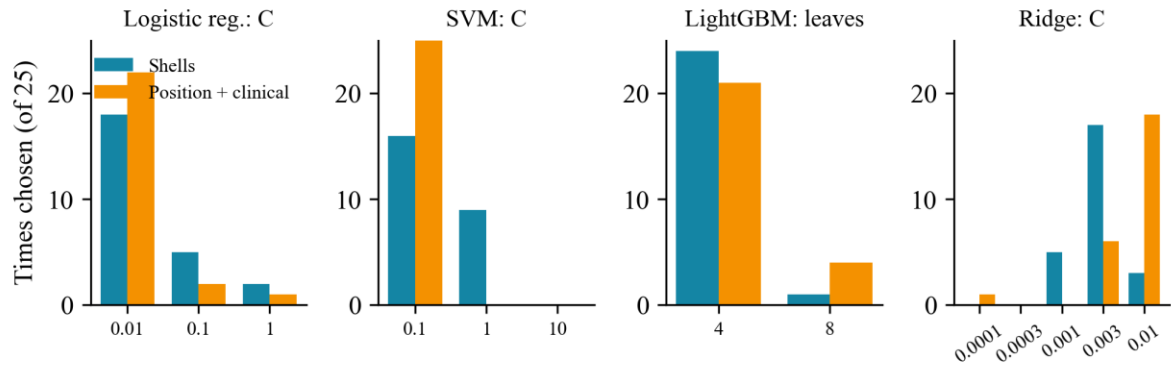


Figure 20: Frequency of each hyperparameter value selected by the inner 3-fold cross-validation in the 25 outer folds

Deep-learning alternatives (E7)

Table 14 lists the larger models that were tested under the same protocol. Frozen FMCIB features [15] reached 0.675. CT-FM [16] reached 0.662 and kept hospital information even after ComBat (detector 0.765), probably because its whole-CT pre-training encodes scanner style. Fine-tuning the last block of FMCIB (35 million parameters) with a gradient-reversal hospital adversary [40] overfitted within a few epochs and scored 0.606, with a detector AUC of 0.815. With about 660 training patients per fold, the networks learned hospital and noise faster than histology.

Approach	Site-balanced AUC	LOSO mean	Hospital detector	Finding
Attention-MIL on FMCIB crop bags	0.669	0.634	n/a	did not beat mean pooling
FMCIB, frozen (3-D ResNet-50)	0.675	0.660	0.263	weaker than handcrafted features
CT-FM, frozen	0.662	0.652	0.765	kept hospital identity
Fine-tuned CNN	0.615	0.644	0.814	overfitted
Fine-tuned CNN + adversary	0.606	0.651	0.815	adversary did not remove hospital
Final model (shells + position)	0.722	0.714	0.54-0.61	best and simplest

Table 14: Final model vs. deep-learning alternatives

Project journey

Figure 21 shows the main AUC at each stage of the project. The points are based on different data and metrics, so the plot tells the story rather than giving a strict comparison. After the progress presentation, the biggest improvements came from more hospitals, a fair metric and better masks, not from bigger models.

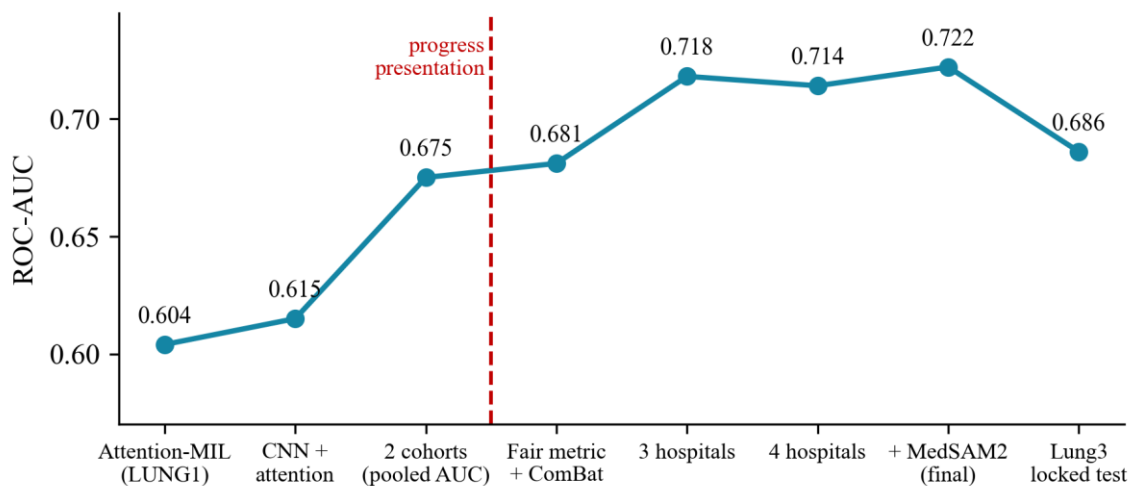


Figure 21: AUC at each stage of the project. The red line marks the progress presentation.

Locked external tests (E8)

Figure 22 and Table 15 show the two pre-registered results. On Lung3 the model achieved 0.686 (95% CI 0.557-0.807), close to the expected LOSO level of 0.714. On TCGA it reached 0.585 (0.462-0.707). Slice thickness was important in both tests: on thin slices (at most 2.5 mm) the AUC was 0.90 for Lung3 and 0.73 for TCGA, while on thick slices it was 0.68 and 0.51. These subgroups are small (18 and 29 thin-slice patients), so they are hints, not proof. TCGA has a further problem: each of its 9 contributing hospitals sent patients with only ADC or only SCC, so in TCGA the hospital is the label. A model that is designed to ignore hospital style cannot use this, and the effective sample size is closer to 9 hospitals than to 84 patients. TCGA tumours were also smaller (median 16 versus 29 ml) and were scanned with more reconstruction kernels (19 versus 10).

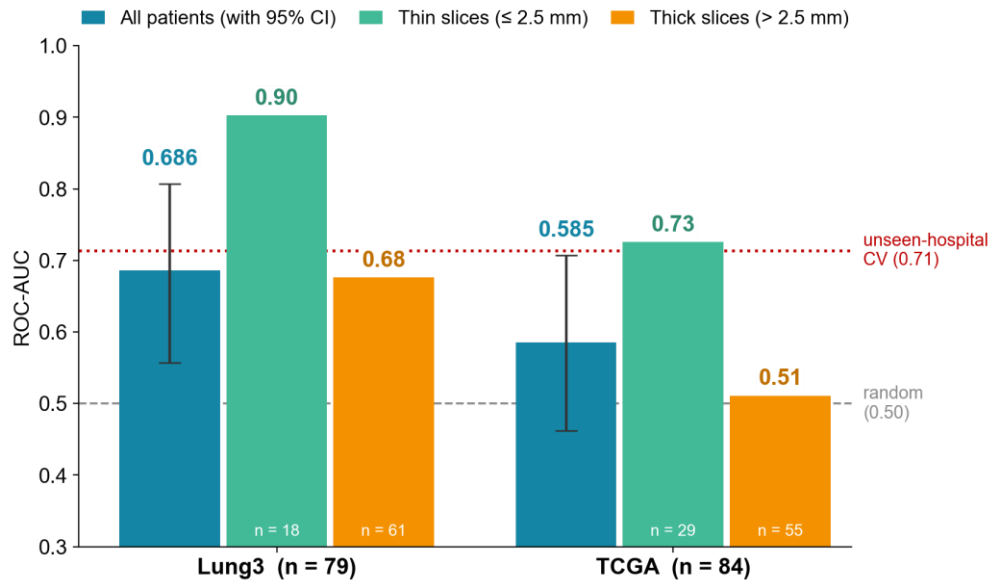


Figure 22: Locked external tests: AUC on all patients with 95% bootstrap CI, and on thin (at most 2.5 mm) and thick slices. The red dotted line is the LOSO level (0.71).

Test set	n (ADC/SCC)	AUC [95% CI]	Se	Sp	BA	Thin slices	Thick slices
Lung3	79 (44/35)	0.686 [0.557, 0.807]	0.636	0.686	0.661	0.903 (n=18)	0.676 (n=61)
TCGA	84 (48/36)	0.585 [0.462, 0.707]	0.562	0.583	0.573	0.726 (n=29)	0.511 (n=55)

Table 15: Locked external test results of the pre-registered model. The thin/thick split was pre-declared for Lung3 and post hoc for TCGA.

Comparison with existing work

Table 16 compares our results with published studies. Scores above 0.8 mainly come from a single hospital or from random splits of pooled data. On truly independent hospitals the published range is about 0.54-0.73. On the same external dataset (Lung3), our locked score of 0.686 is higher than the 0.60 reported by Chaunzwa *et al.*, while Song *et al.* reported 0.80 on Lung3 and 0.82 on TCGA. Their numbers were obtained with expert tumour masks, by selecting the best of 130 models on the test sets, and with hospitals mixed randomly in training, so the two results were produced under different rules.

Study	Data	How it was tested	AUC	Remarks
Zhu 2018 [9]	129 pts, 1 hospital	split inside one hospital	0.91	small, no external test
Pasini 2023 [11]	466 pts incl. LUNG1 + Radiogenomics	random 80/20 split	Acc 0.77 to 0.59	drop after ComBat
Yang 2021 [10]	645 pts, 3 centres	train one centre, test others	0.54-0.64	0.78 with random mixing
Chaunzwa 2021 [13]	311 pts, 1 hospital	external Lung3 (49 pts)	0.71 to 0.60	deep CNN
Chen 2023 [14]	402 pts, TCIA	external test (78 pts)	0.84 to 0.73	multi-task CNN
Song 2023 [12]	868 pts, 8 TCIA sets	internal + TCGA (97) + Lung3 (71)	0.82 / 0.82 / 0.80	best of 130 models, expert masks

Study	Data	How it was tested	AUC	Remarks
This work	1041 pts, 4 hospitals	site-balanced CV, LOSO, 2 locked tests	0.72 / 0.71 / 0.69 / 0.59	fair metric, automatic masks, pre-registered

Table 16: Comparison with existing ADC/SCC studies (internal to external where two values are given)

5.3.4 Interpretation and Conclusions on Data

- Pooled AUC can be deceptive. Multi-hospital ADC/SCC data show the hospital alone accounts for a pooled AUC of approximately 0.71. Any multi-centre result should be validated with a site-balanced metric or per hospital AUCs.
- There is a real CT signal. Under the fair metric the model achieves 0.722, keeps 0.714 on unseen hospitals and reaches 0.686 on the locked Lung3 test. The signal comes mainly from tumour position, size and the texture of the tumour border, which agrees with known clinical tendencies.
- Harmonisation makes results more reliable rather than higher. ComBat removed most of the scanner fingerprint with almost no change in the AUC.
- Data diversity was more important than model complexity. Adding hospitals and better masks helped; attention, CNNs, foundation models and adversarial training did not.
- The main limit is slice thickness. A large part of the external drop is explained by the fact that texture learned mostly on thin CT does not transfer to 5 mm CT. The model should only be applied on thin-slice CT.

5.4 Limitations of Tools (PO(e))

Table 17 summarises the main technical limitations of the tools we used and their impact on the results.

Tool	Limitation	Effect / mitigation
PyRadiomics	texture depends on voxel size, bin width and slice thickness	fixed 2 mm grid and 20 HU bins; still unreliable on 5 mm CT
Resampling (SimpleITK)	cannot recover detail lost between thick slices	5 mm scans become blurred along z; reported as main failure mode
TotalSegmentator	picks the largest nodule; can miss or choose the wrong lesion	nodule found in 55/60 checks, correct lesion in about 80%; about 1 in 5 external masks may be wrong
MedSAM2	depends on the box; Dice 0.74 with loose boxes; hole filling not available on Windows	box enlarged by only 3 mm; validated on 343 expert outlines
ComBat	assumes a location-scale shift; may remove real signal when class mixes differ; needs enough patients per site	fitted per fold; new sites mapped label-free; balanced variants tested
Classifiers (LR, SVM, LightGBM)	small data, noisy labels	strong regularisation, ensembles, nested tuning
Bootstrap CI	wide intervals at n of about 80	external CIs are about plus or minus 0.12; reported in full
Hardware (4 GB GPU)	limits network size and batch size	frozen backbones; deep models restricted to last-block fine-tuning
Public labels	histology from records (NLST codes, TCGA collection)	strict inclusion rules; mixed types excluded

Table 17: Limitations of the tools used

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

A dependable CT-based prediction of cancer type could assist in reducing delays and repeat biopsy; this is especially important in settings where pathology services are limited. In contrast, the risk of over-trust is that a confident-looking score is treated as a diagnosis. Communication to doctors must emphasise that the tool is complementary to, not a replacement for biopsy. As there are no South Asian patients in our data, local validation is required to prevent a tool from performing worse for the population it is supposed to help.

5.5.2 Assessment of Health and Safety Issues

The pipeline is non-invasive and uses existing scans, so there is no direct physical risk. The indirect risk is a wrong prediction. A false ADC call could delay the correct workup of an SCC patient, and vice versa. With the current accuracy, clinical decisions cannot be based on the model. Before any clinical use it would require prospective testing, a radiologist check of the detected nodule, and rules for when the model must not be used (for example 5 mm CT).

5.5.3 Assessment of Legal Issues

All datasets are public and de-identified, and were used under their licences: TCIA collections under CC BY 3.0/4.0 or CC BY-NC 3.0 (non-commercial, e.g. Lung3) and NLST data through the NCI Imaging Data Commons. The software tools are available under open-source licences (Apache 2.0, BSD or MIT). A clinical version would be software as a medical device and would require regulatory approval (for example by the Directorate General of Drug Administration in Bangladesh, FDA in the USA or CE marking in Europe) and compliance with patient data protection rules when used on hospital data.

5.6 Sustainability Evaluation (PO(g))

Technical sustainability

All configurations, random seeds, splits, feature names and package versions are saved. Extraction is resumable and all experiments can be rerun from the saved features. The model is small and can be retrained in minutes when a new hospital joins the network.

Economic sustainability

The pipeline uses freely available open-source software and can be run on a mid-range laptop, so the hospital does not need any special hardware (Section 8).

Environmental sustainability

The energy consumption is low (approx. 10-15 kWh for the whole project) because we did not train large networks and reused public data.

Long-term use

Label-free ComBat mapping allows for the addition of new hospitals without their labels, and the locked-test protocol can be repeated whenever the model is updated.

5.7 Ethical Issues (PO(h))

The following ethical principles were used:

- Privacy: Only de-identified public data were used; no attempts were made to identify any

patient, and the data were stored locally and not redistributed.

- Honest evaluation: each fitted step was confined to the training folds, external tests were pre-registered, frozen with a SHA-256 hash and scored once, and post hoc analyses are clearly labelled.
- Reporting of negative results: failed approaches (attention-MIL, foundation models, adversarial fine-tuning) and the weak TCGA result are reported in full, rather than hidden.
- Fairness: site-balanced metric weights all hospitals equally and results are presented on a per-hospital basis so poor performance in one group is visible.
- No overclaiming: the model is presented as a research decision-support tool rather than a diagnostic device.
- Credit: all datasets, tools and published methods are referenced.

One ethical challenge was the incentive to modify the model after observing the first external outcome (TCGA). We did not change the model, but the second test (Lung3) was pre-registered with the same frozen model, and the thin/thick analysis was pre-registered for it.

6 Reflection on Individual and Team work (PO(i))

This was an individual project: there is one person in Group 13. The reflection below thus describes how the work was organised, and how the feedback from the instructors was used.

6.1 Individual Contribution of Each Member

Member	Contribution
Rahib Mahasin (2106119)	Topic selection and literature review; data download and preprocessing; segmentation and feature extraction; design of the fair metric and validation; all model experiments; locked external tests; figures, presentation slides and this report.

Table 18: Individual contribution

6.2 Mode of TeamWork

Experiments were planned on a weekly basis with a short list of goals and a written log of all experiments (phase reports with commands, results and warnings). The instructor feedback on the progress presentation in week 10 (more data, external validation, clearer evaluation) directly shaped the second half of the project. Decisions that might bias the results, including the choice of the final model and the rules for the external test, were documented before the data were analysed.

6.3 Diversity Statement of Team

Diversity was in the sources used rather than the team members: data from three countries and four clinical settings (radiotherapy, surgery, diagnosis and screening), methods from medical imaging, statistics and deep learning, and published work from many groups. We chose a site-balanced evaluation to ensure that smaller hospitals and minority classes are represented fairly in the result.

6.4 Log Book of Project Implementation

The project started at the mid-term break (after Week 7) and continued until the final demonstration in Week 14. The weekly log is summarised in Table 19.

Date	Milestone achieved	Individual Role	Team Role	Comments
Mid-break	Topic chosen; literature review; LUNG1 data and ROI products checked; method plan locked; 29,184 local patches extracted	Literature, data preparation	Sole member	planning and setup
Week 8	Mean vs attention MIL on Gradient ROI (0.604); GTV + rim bags, 37,464 patches (0.577)	Model training	Sole member	attention unstable
Week 9	mRMR/LASSO selection (0.593); ResNet-18 patch embeddings (0.615); CLAM loss; global radiomics	Experiments	Sole member	deep features first helped attention
Week 10	Radiogenomics added (343 pts, 0.675); hospital detector 0.998; progress presentation	Data, presentation	Sole member	feedback: more data, external test
Week 11	Shortcut measured; site-balanced AUC; ComBat; shells and position features (0.681)	Method redesign	Sole member	first honest result
Week 12	Lung-PET-CT-Dx (0.718) and NLST added: 1041 pts, LOSO 0.704; TCGA downloaded and locked	Data scale-up	Sole member	about 80 GB downloaded
Week 13	MedSAM2 masks (0.722); CT-FM and adversarial tests; locked TCGA (0.585) and Lung3 (0.686)	Segmentation, testing	Sole member	pre-registered, scored once
Week 14	Slides, final demonstration and report	Communication	Sole member	project finished

Table 19: Log book of project implementation

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

Can a CT scan tell which type of lung cancer a patient has? Students at BUET have created a computer program that looks at a routine chest CT scan and estimates whether a lung tumour is an adenocarcinoma or a squamous cell carcinoma, the two main types, which are treated differently. The program automatically detects and outlines the tumour, measures its position and the texture around it, and removes differences caused by different scanners. It was tested on 1041 patients from hospitals in three countries and then on two new datasets that were kept hidden until the end, and it was right in about seven out of ten comparisons. It is a research tool that could one day support, not replace, biopsy.

7.2 User Manual

The following steps run the frozen model on a new CT scan. Commands are run from within the project folder, using the Python environments described in Section 4.

1. Set up. Create the three environments (main, MedSAM2, TotalSegmentator) from the saved requirement files and download the MedSAM2 checkpoint.
2. Get the scan ready. Organise CT DICOM series of each patient in separate folders. The scan should be a pre-treatment chest CT, preferably with slices of 2.5 mm or less.
3. Convert and resample. Run the build script to convert DICOM to NIfTI and resample to 2 mm.
4. Find the tumour. Run the detection script (TotalSegmentator lung nodules) and check the detected box in the saved preview image. If the wrong nodule was selected, replace the box by hand.
5. Outline the tumour. Run the MedSAM2 script with the saved settings. It writes one mask per patient.

6. Extract features. Run the feature script for radiomics (shells) and position; age and sex are read from the DICOM header.
7. Predict. Run the prediction script. It fits the frozen model on the 1041 development patients, maps the new scans as a new site with label-free ComBat, and writes one score per patient.
8. Read the result. A score above 0 suggests ADC and a score below 0 suggests SCC; scores near 0 are uncertain. The result should be interpreted by a physician together with other clinical information.

7.3 Github Link

The complete source code, configuration files, documentation, this report and the presentation slides can be found in the GitHub repository below. The main parts of the code are also listed in Section 3.7.

<https://github.com/raahiibbb/ADC-SCC-CT-Radiomics>

8 Project Management and Cost Analysis (PO(k))

The project is software only so the costs are in computing, data transfer and time rather than hardware components. The figures below are estimates in Bangladeshi Taka (BDT).

8.1 Bill of Materials

Item	Quantity / use	Cost (BDT)
Laptop with RTX 3050 GPU (already owned)	share of value used over 2 months (about 90,000 BDT over 4 years)	3,750
Storage for data (part of a 1 TB disk)	about 150 GB	900
Internet data	about 90 GB downloaded over 2 months	2,500
Electricity	about 15 kWh at about 12 BDT/kWh	180
Software (Python, PyRadiomics, scikit-learn, LightGBM, MedSAM2, TotalSegmentator)	open source	0
Datasets (TCIA, IDC)	public	0
Total		about 7,330

Table 20: Estimated bill of materials

8.2 Calculation of Per Unit Cost of Prototype

The trained pipeline is the prototype. Its development cost is the total of Table 20, about 7,330 BDT, plus the student's time (about 8 weeks). The compute part is small: development runs used about 100-150 GPU-hours of laptop time at about 0.1 kW, i.e. 10-15 kWh.

8.3 Calculation of Per Unit Cost of Mass-Produced Unit

For a deployed tool, the relevant unit is one patient analysed. Measured times on the laptop were about 2.3 minutes per patient for nodule detection (3.2 hours for 85 TCGA scans), about 2 seconds for MedSAM2 (38 minutes for 1041 patients), about 2 seconds for feature extraction and less than a second for prediction, so about 2.5 minutes in total. With a power draw of 0.1 kW and electricity at 12 BDT/kWh, the cost per patient is

$$C_{\text{patient}} = 0.1 \times \underbrace{\frac{2.5}{60}}_{\text{energy}} \times 12 + \underbrace{\frac{150000}{5 \times 20000}}_{\text{workstation}} \approx 0.05 + 1.5 \approx 1.6 \text{ BDT} \quad (27)$$

assuming a dedicated 150,000 BDT workstation used for 20,000 scans per year for five years. Even with the cost of maintenance and staff time, the cost per patient is a tiny fraction of the cost of a biopsy.

8.4 Timeline of Project Implementation

Figure 23 shows the timeline as a Gantt chart. The work was carried out over eight weeks: the mid-term break and Weeks 8-14. The progress presentation took place in Week 10 and the final demonstration in Week 14.

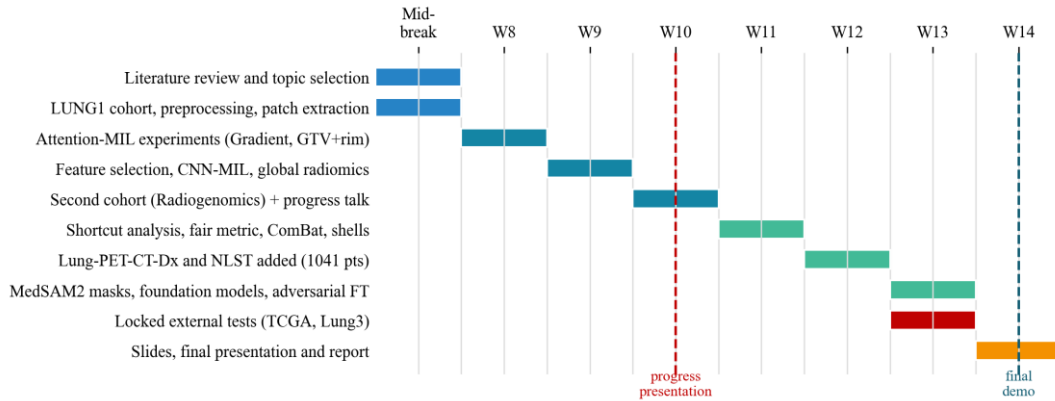


Figure 23: Gantt chart of the project.

9 Future Work (PO(I))

The project is complete but the results point to distinct next steps:

- Robustness to thick slices. Train on thin scans artificially blurred to 5 mm, or use features that are less sensitive to slice thickness, because 5 mm CT was the main failure mode.
- Larger external test. Test on a multi-hospital dataset with both ADC and SCC at each hospital, so that per-hospital AUCs can be computed, ideally including Bangladeshi hospitals.
- Human check of the detected nodule. A short confirmation of the automatic box by a radiologist would remove most of the wrong-lesion errors (about 1 in 5).
- Clinical and PET data. PET uptake and smoking history (if available) may provide additional information that is not contained in CT texture.
- Better use of deep models. Foundation models could be revisited with harmonisation built into training, or with much larger multi-hospital data.
- Prospective evaluation. Before any clinical use, the frozen model should be tested prospectively on new patients with the same locked protocol.

10 References

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